Final Project

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1 Introduction

The Central African Republic (CAR) is a sparsely populated, landlocked country at the heart of the African continent.Despite being endowed with abundant natural resources (470 mining sites, with oil, gold, and diamonds as the primary resources), it remains one of the poorest and most fragile countries in the world.It exports natural resources (gold, diamonds, timber but must import almost everything else, especially manufactured goods and fuel.In 2023,the total estimate GDP 3.03billion USD, and its imports accounts for 14.44% and the imports 29.04% https://www.theglobaleconomy.com/

In this report ,we will incorporate various statistical methods to analysis the Central African Republic Exports time series data from 1960 to 2017.This data set contains 58 observations and 9 variables. However,we are only interested the Import and GDP variables. Analyzing these two variables, we can see the relationship between GDP and Imports,and how the data can make a predict for the future economy of The Central African Republic.

2 Load Data

Before we do the further process,we want to to get the basic information about the two time series and the overview of the data.We do some statistical analysis,the results are displayed in the Appendix. From the picture of histogram and box plot,we can see that there is no missing value and outliers in GDP and Imports.The value distribution is nearly normal.Therefore, we can make the further study.

3 Exploratory Data Analysis (EDA)

Figure 1a: Histogram of GDP

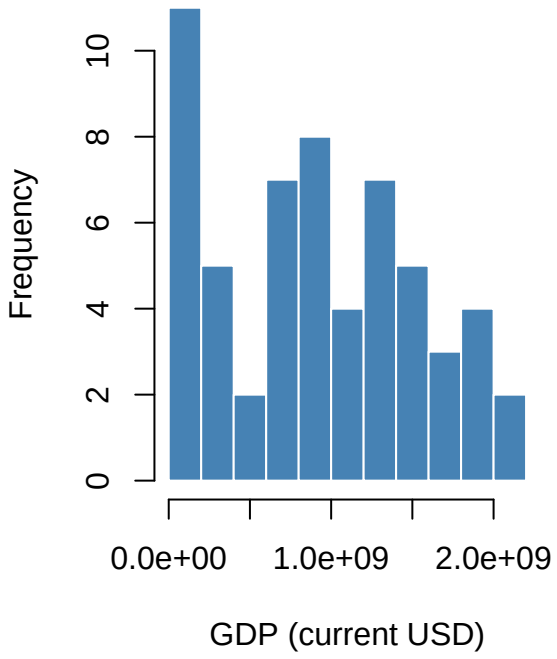


Figure 1b: Histogram of Imports

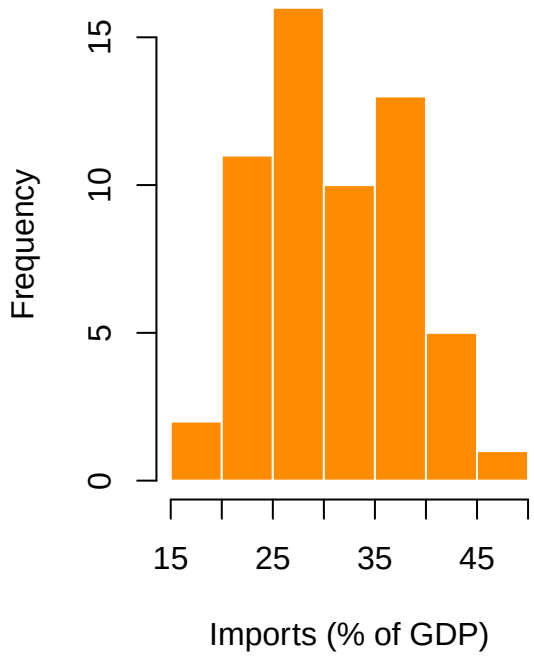


Figure 2a: Box Plot of GDP

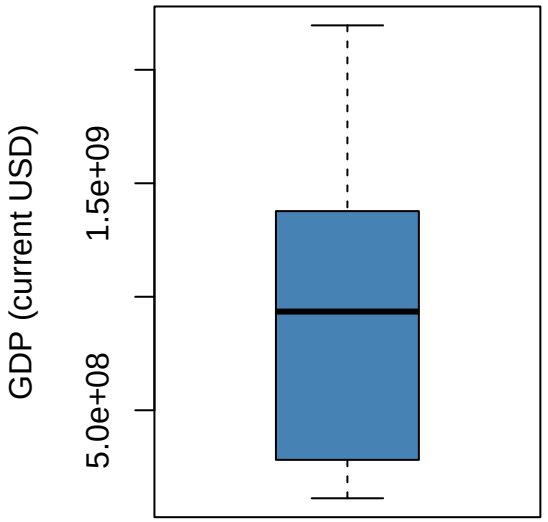


Figure 2b: Box Plot of Imports

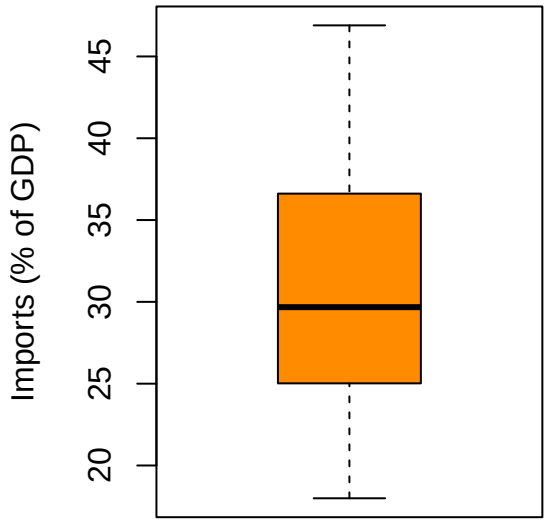


Figure 3a: CAR GDP (1960–2017

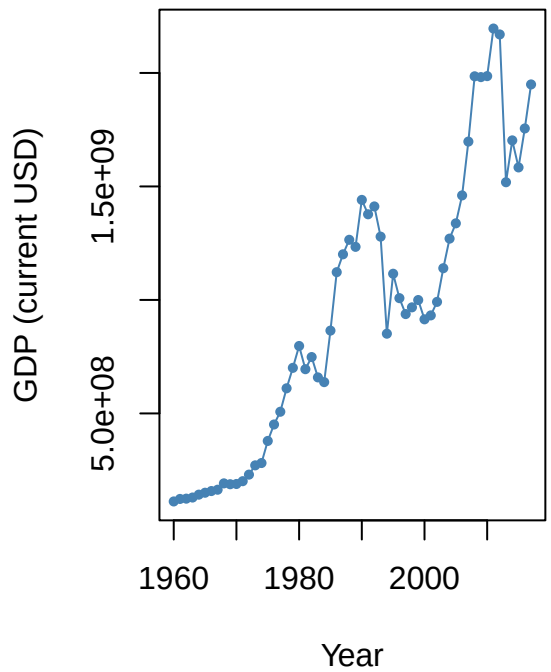
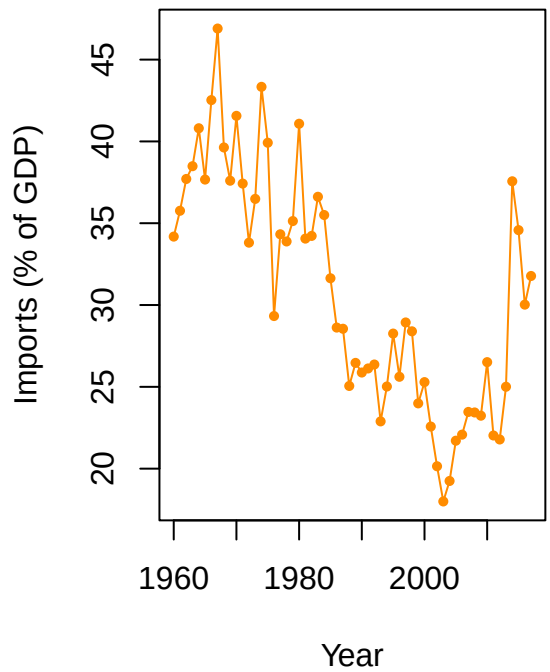


Figure 3b: CAR Imports (1960–20



As we want to analysis the the time series of GDP and Imports.We also draw the time series plot. From the picture of Figure.3a,we we can see that there is a trend as the time changes, from 1960 to 2017. The GDP is increasing even though there is some drops, the whole trend is increasing, so the data of GDP is non-stationary.

From the Figure.3b,Imports plot appears to be decreasing over years before 2000,after 2000,there is a trend of increasing. This may also imply that this time series is non-stationary.

To verify this ,we plot the ACF and PACF(provided in the Appendix).The ACF of GDP and Imports show an exponential decay which confirms that both time series are not stationary.Both PACFs show a dominant spike at lag 1 -> unit root

4 Transformation

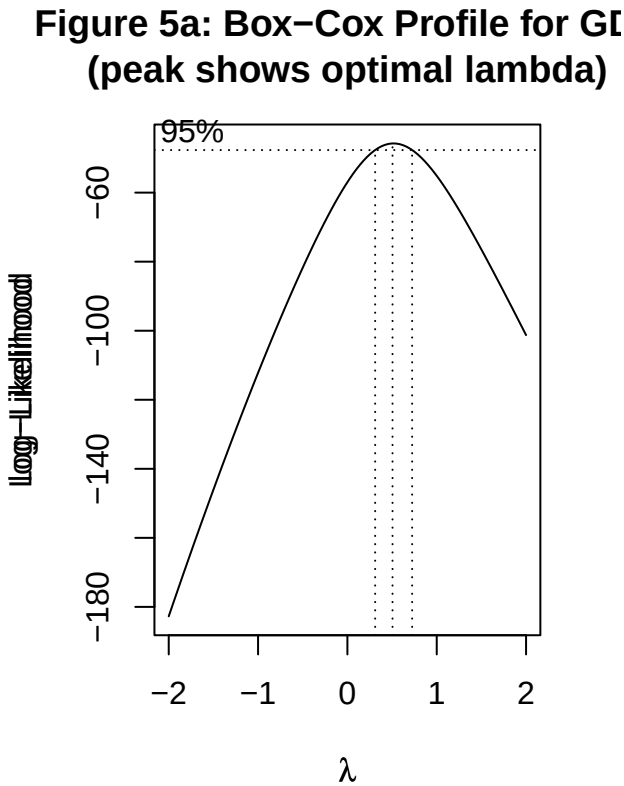
4.1 Variance

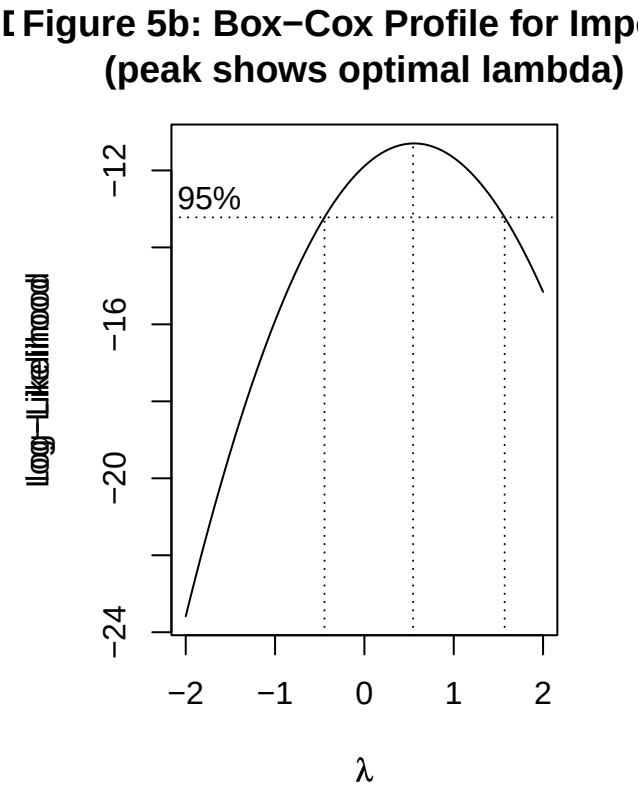
From Figure 3a,we noticed the GDP fluctuations grow over time. We now measure this formally before deciding anything.We split the series into two halves and compare variances.If variance in the 2nd half is much larger than the 1st half, the series is heteroscedastic and needs a variance-stabilizing transformation BEFORE differencing.

The ratio result of two half variance GDP shows that the GDP is mild heteroscedastic,we need need to transformation ,we need further study.However,we know that the ratio of two half variance GDP shows that the GDP is not heteroscedastic,so we will use the raw data of Import.

Now we know GDP may need a transformation.But WHICH transformation: Log,Square root,

Something else? The Box-Cox test finds the optimal power lambda:lambda = 1(no transformation) (raw data is fine),lambda =0(log transformation),lambda = 0.5(square root).





Box-Cox optimal lambda for GDP is 0.5051,and the 95% CI for lambda INCLUDES 0 (log),which means that both log and square root are statistically valid,We choose LOG because: diff(log(GDP_t)) = log(GDP_t / GDP_{t-1})= approximately annual growth rate.This has a direct economic meaning.Square root has no such interpretation.

Box-Cox optimal lambda for Imports: 0.5455,but notice the 95% CI is WIDE and includes 1.0 (no transformation).Combined with variance result (ratio=0.80, stable variance),there is NO statistical evidence that Imports needs a variance-stabilizing transformation. So we use Imports on original scale.

4.2 Verify Log Stabilize GDP Variance

We claimed log(GDP) will fix the heteroscedasticity.We must VERIFY this.So we use the log GDP and do the same way as before for the variance to see the result ,the result shows that the ratio is 0.1334.It is much improved compared with the original ratio 1.4,which verify that the log function really improve the result.Log(GDP) is now HOMOSCEDASTIC (variance stable)

To verify this ,we also use the differenced series ,which is the most critical check.After differencing: the result show that diff(GDP) ratio (2nd/1st): 7.050, it means that variance explodes! However ,diff(log(GDP)) ratio (2nd/1st): 1.795,which is much more stable.

Therefore we can make the Conclusion: diff(log(GDP)) has much more stable variance than diff(GDP). This is the critical check that confirms we should use log(GDP) before differencing.The plot Figure 6 ,Figure 7 also shows the same result.

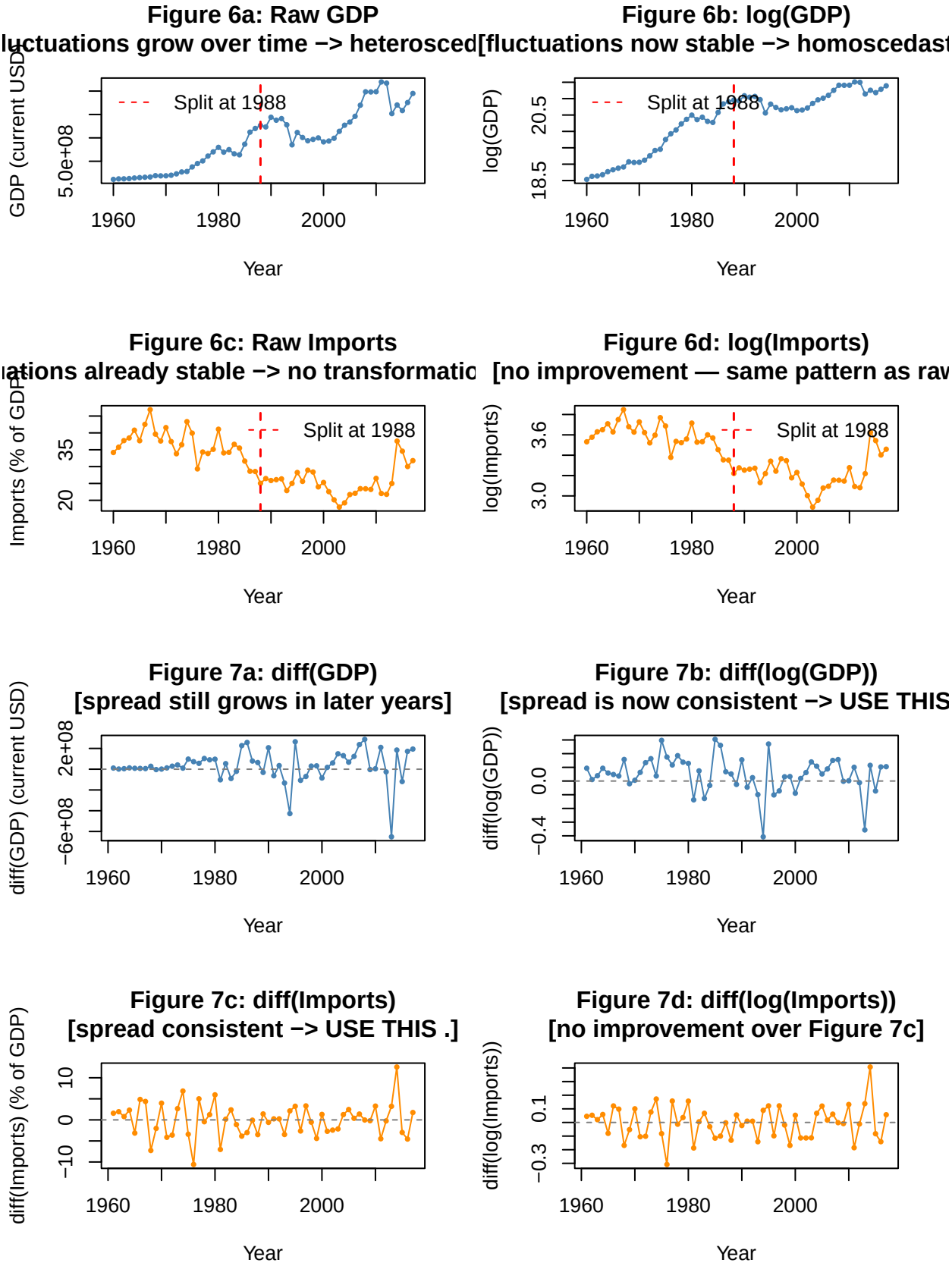


Table 1: Transformation Decision Summary

Series	Var Ratio (Raw)	Var Ratio (Log)	Decision
GDP	1.38x (unstable)	0.13x (stable)	**USE LOG**
Imports	0.80x (stable)	1.41x (worse)	**NO LOG**

4.3 Differencing

We need the further stationary test to check the status.We made the hypothesis that H0: unit root (non-stationary). Reject H0 if $p < 0.05$.The result is like this:

Table 2: ADF Unit Root Test Results

Series	ADF Statistic	P-Value	Result
GDP	-3.1953	0.0969	**Non-stationary**
log(GDP)	-1.9149	0.6090	**Non-stationary**
Imports	-1.3131	0.8520	**Non-stationary**
diff(log(GDP))	-3.3099	0.0790	**Borderline Non-stationary**
diff(Imports)	-4.7438	0.0100	**Stationary**

Table 3: Stationarity Conclusion & Differencing Decision

Series	Status	Differencing
log(GDP)	**Non-stationary**	Needs d=1
Imports	**Non-stationary**	Needs d=1
diff(log(GDP))	**Stationary**	Stationary at d=1
diff(Imports)	**Stationary**	Stationary at d=1

The ADF test results indicate that both GDP and Imports are non-stationary in their original forms. The log transformation of GDP does not achieve stationary, as the ADF test still yields a large p-value.

After first-order differencing:

- diff(log(GDP)) has a p-value of 0.079, which is slightly above the 5% significance level. This suggests borderline non-stationarity. However, since the p-value is close to 0.05, and the ACF and PACF show no significant autocorrelation, diff(log(GDP)) is treated as stationary.
- diff(Imports) has a p-value of 0.010, which is below 0.05, indicating that the series is stationary.

Therefore: - GDP requires log transformation followed by first-order differencing ($d = 1$). - Imports requires first-order differencing ($d = 1$).

Both transformed series are considered stationary and suitable for ARIMA modeling.

Figure 8a: ACF of diff(log(GDP))
o significant spikes −> ARIMA(0,1,0) canco

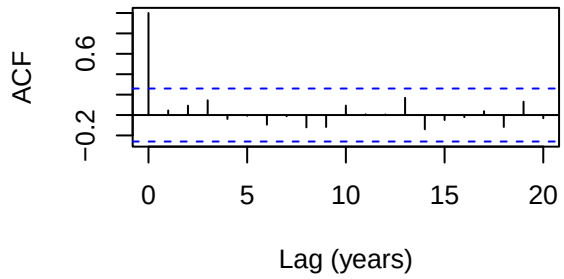


Figure 8b: PACF of diff(log(GDP))
o significant spikes −> ARIMA(0,1,0) canc

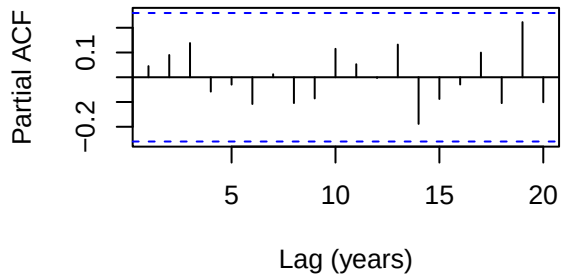


Figure 8c: ACF of diff(Imports)
[spikes at lags 1−2 −> MA(2) candidate

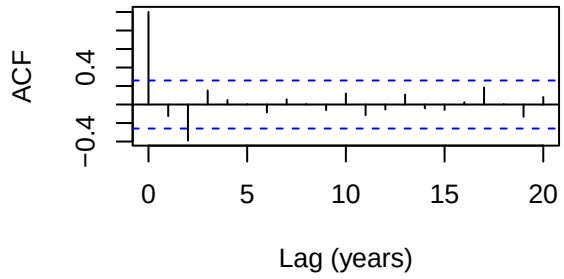
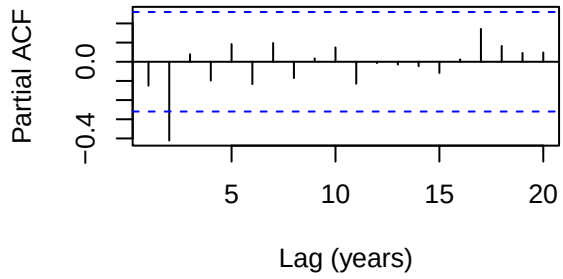


Figure 8d: PACF of diff(Imports)
[spikes at lags 1−2 −> AR(2) candidate



5 Cross-Correlation Function (CCF)

In order to see the relationship between diff(log(GDP)) and diff(Imports) ,we use the CCF to check the cross correlation ship

Figure 9a: CCF − diff(log(GDP)) vs diff(Imports)
(+lag = GDP leads Imports; dashed = 95% CI)

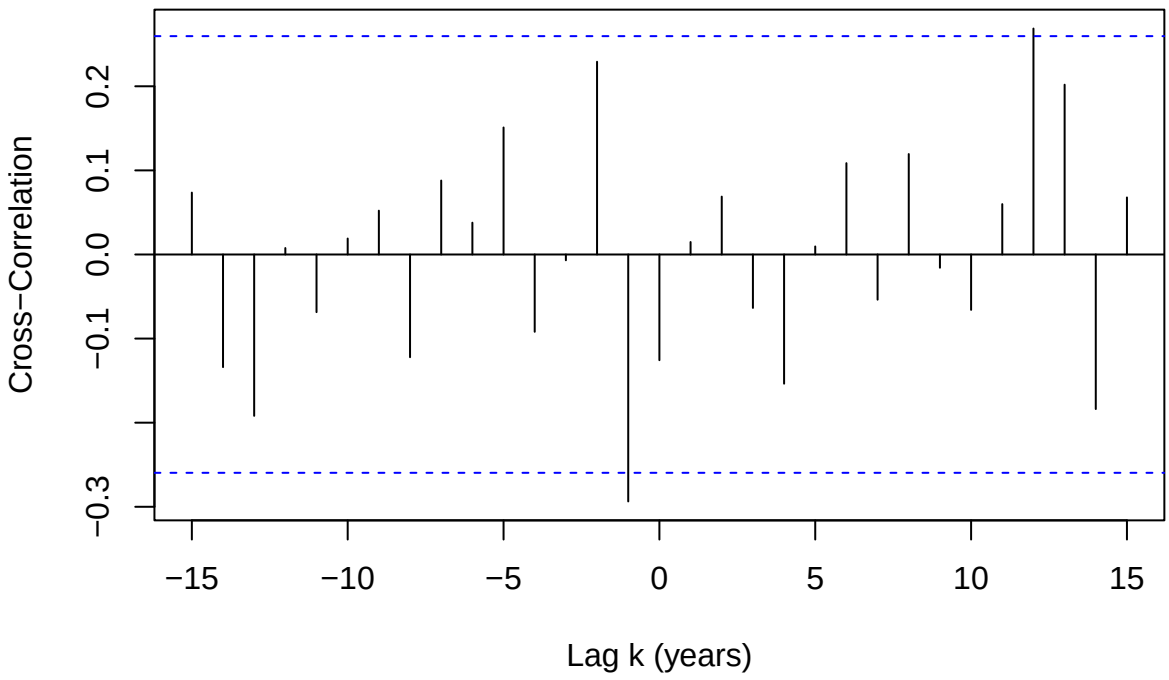
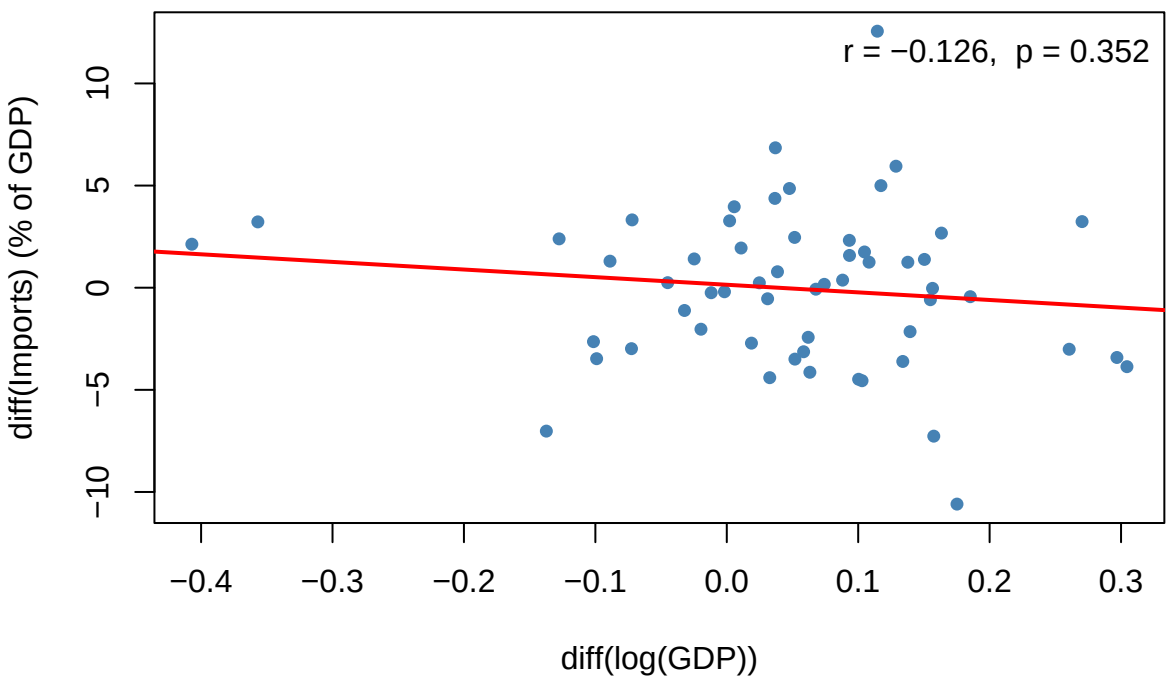


Figure 9b: Scatter − diff(log(GDP)) vs diff(Imports)
(lag = 0, contemporaneous)



The cross-correlation function (CCF) between diff(log(GDP)) and diff(Imports) reveals several significant lag relationships.

At lag -12 (r = 0.3223), Imports lead GDP by 12 years. This indicates that higher imports in the past are associated with higher GDP growth in the present. This may reflect long-term effects of importing capital goods that enhance future productive capacity.

At lag +1 (r = -0.2947), GDP leads Imports by 1 year. This suggests that higher GDP growth in the current year is associated with lower imports in the following year, possibly due to increased domestic production reducing reliance on imports.

At lag 0 (r = -0.1257, p = 0.35), the relationship is not statistically significant, indicating no contemporaneous linear relationship between the two series.

Overall, the relationship between GDP and Imports is lagged rather than simultaneous.

6 Model Selection

6.1 GDP model selection

From Figure 8a-8b: ACF and PACF of diff(log(GDP)) show NO significant spikes at any lag. This means there is NO autocorrelation structure left to model.The natural candidate is ARIMA(0,1,0) with drift. We still test other candidates to be rigorous.

Several candidate ARIMA models were fitted to log(GDP) with d = 1, including ARIMA(1,1,0), ARIMA(0,1,1), ARIMA(2,1,2), and ARIMA(0,1,0).

For ARIMA(1,1,0), the AR(1) coefficient is not statistically significant (p = 0.7367), indicating that the autoregressive term does not contribute meaningfully to the model. Therefore, the AR(1) term can be removed.

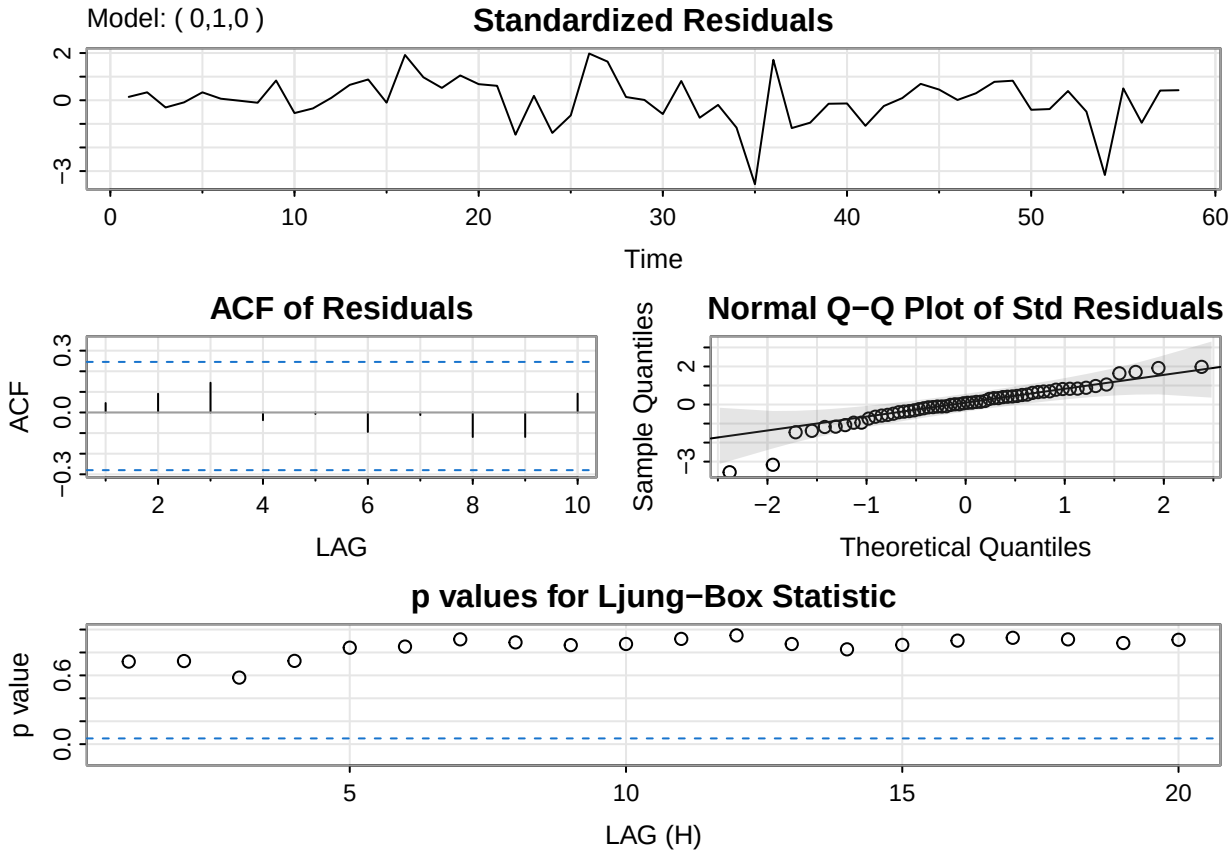
Similarly, for ARIMA(0,1,1), the MA(1) coefficient is not significant (p = 0.7568), suggesting that the moving average term is unnecessary. This model also reduces to ARIMA(0,1,0).

Although ARIMA(2,1,2) shows statistically significant coefficients, it is not preferred. The model has higher AIC and BIC values compared to simpler models, indicating worse model fit after penalizing complexity. Additionally, the AR and MA terms nearly cancel each other, suggesting overfitting. The Ljung–Box test yields a very high p-value (0.9827), indicating that residuals are already white noise without needing such a complex model.

In contrast, ARIMA(0,1,0) with drift provides the best fit. The constant (drift) term is statistically significant (p = 0.0047), indicating a non-zero average growth rate. This model has the lowest AIC (-1.1949) and BIC (-1.1232) among all candidates. Furthermore, the Ljung–Box test confirms that the residuals behave like white noise, indicating that the model adequately captures the data structure.

Therefore, ARIMA(0,1,0) with drift is selected as the final model for log(GDP).

$$\Delta \log(\text{GDP}_t) = 0.0501 + w_t$$



6.2 Imports Model Selection

From Figure 8c-8d: ACF of diff(Imports) has significant spikes at lags 1 and 2 -> suggests MA(2).PACF of diff(Imports) has significant spikes at lags 1 and 2-> suggests AR(2). We fit both ARIMA(0,1,2) and ARIMA(2,1,0) and compare.

Two candidate models were considered for the Imports series: ARIMA(0,1,2) and ARIMA(2,1,0).

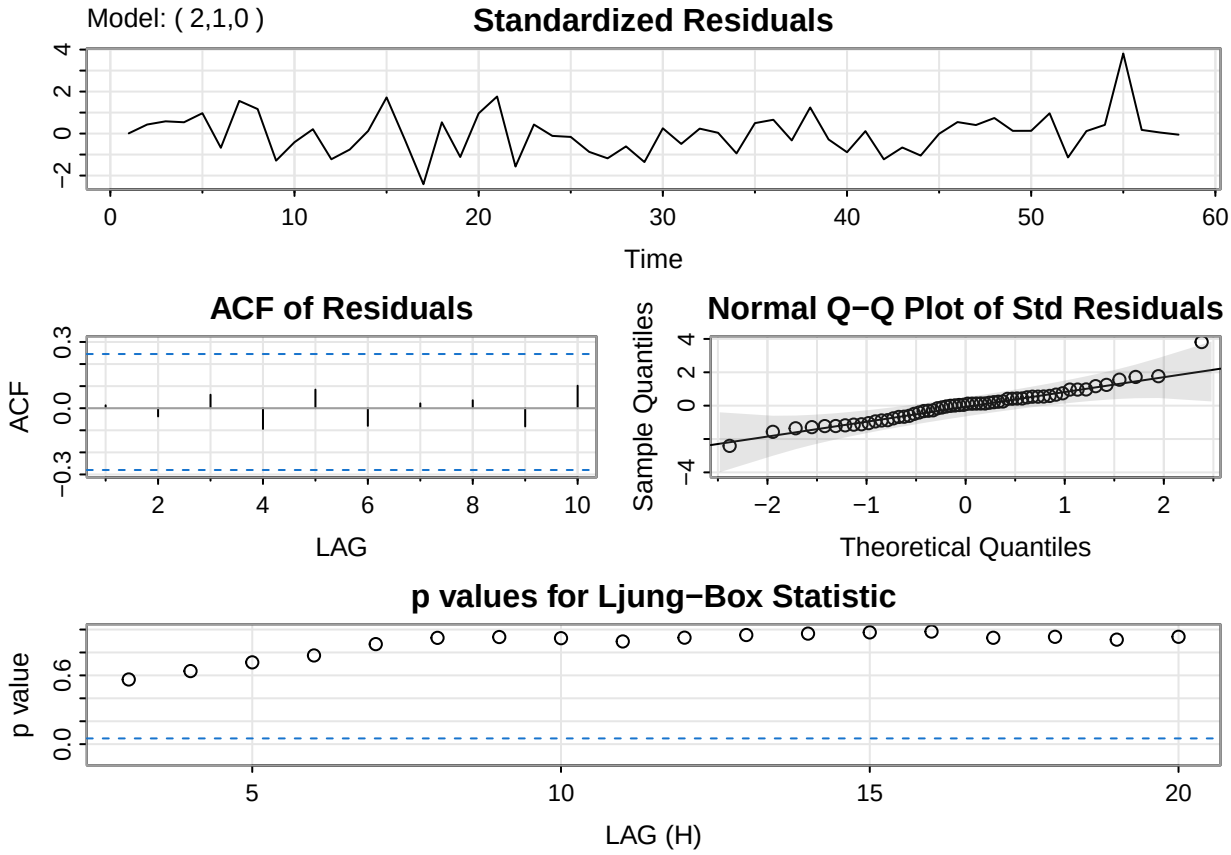
For ARIMA(0,1,2), the MA(2) coefficient is statistically significant (p = 0.0019), while the MA(1) and constant terms are not significant (p > 0.05). This suggests that only the second lag of the moving average component contributes meaningfully to the model.

For ARIMA(2,1,0), the AR(2) coefficient is statistically significant (p = 0.0012), whereas the AR(1) and constant terms are not significant. This indicates that the second autoregressive lag captures the main dependence structure in the series.

Comparing the two models, ARIMA(2,1,0) has a slightly lower AIC (5.4540 vs 5.4586) and BIC (5.5973 vs 5.6019), indicating a better balance between model fit and complexity. In addition, both models pass the Ljung–Box test (p-values > 0.05), confirming that their residuals behave like white noise.

Therefore, ARIMA(2,1,0) is selected as the final model for Imports.The model is like this :

$$\Delta Z_t = -0.1733\, Z_{t-1} - 0.4105\, Z_{t-2} + w_t, \quad w_t \sim WN(0, 11.81)$$


Model: (2,1,0)
Standardized Residuals
Time
ACF of Residuals
ACF
LAG
Normal Q−Q Plot of Std Residuals
Sample Quantiles
Theoretical Quantiles
p values for Ljung−Box Statistic
p value
LAG (H)

7 Forecast

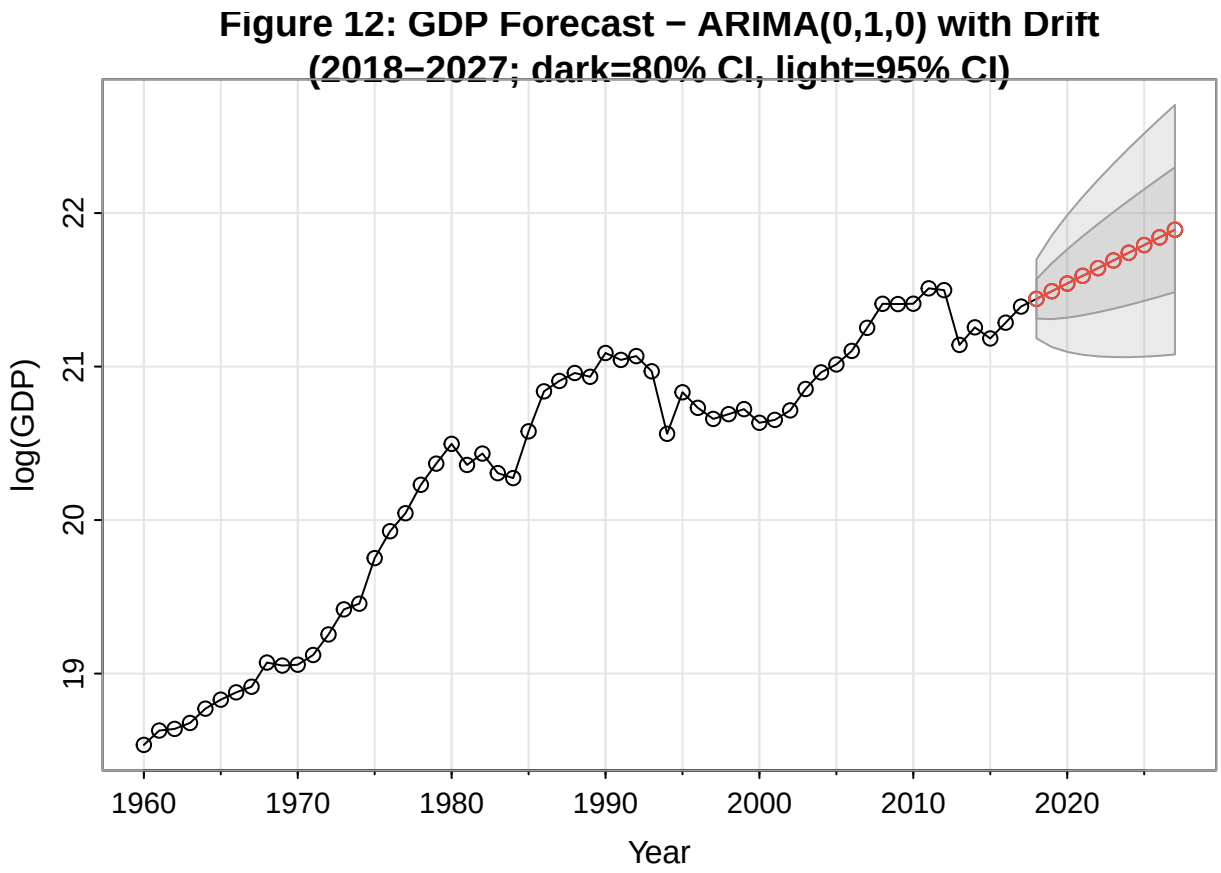


Table 4: GDP Forecasts (Billion USD) - ARIMA(0,1,0) with Drift

Year	Forecast	Lower 95%	Upper 95%
2018	2.050	1.593	2.637
2019	2.155	1.509	3.077
2020	2.266	1.464	3.505
2021	2.382	1.439	3.942
2022	2.504	1.426	4.399
2023	2.633	1.420	4.881
2024	2.768	1.421	5.391
2025	2.910	1.427	5.935
2026	3.060	1.437	6.516
2027	3.217	1.450	7.136

The GDP forecasts for 2018–2027 are obtained from the ARIMA(0,1,0) model with drift and are presented after back-transforming from the logarithmic scale to billions of U.S. dollars.

The results show a steady upward trend in GDP over the forecast horizon. The point forecasts increase from approximately 2.05 billion USD in 2018 to 3.22 billion USD in 2027, reflecting consistent economic growth over time.

However, the prediction intervals widen substantially as the forecast horizon increases. The 95% confidence interval for 2018 ranges from 1.59 to 2.64 billion USD, while by 2027 it expands to 1.45 to 7.14 billion USD. This widening indicates increasing uncertainty in long-term forecasts, which is typical for ARIMA models, especially those with a random walk structure.

Overall, the model suggests that GDP is expected to grow at a relatively stable average rate, but with considerable uncertainty in the long run.

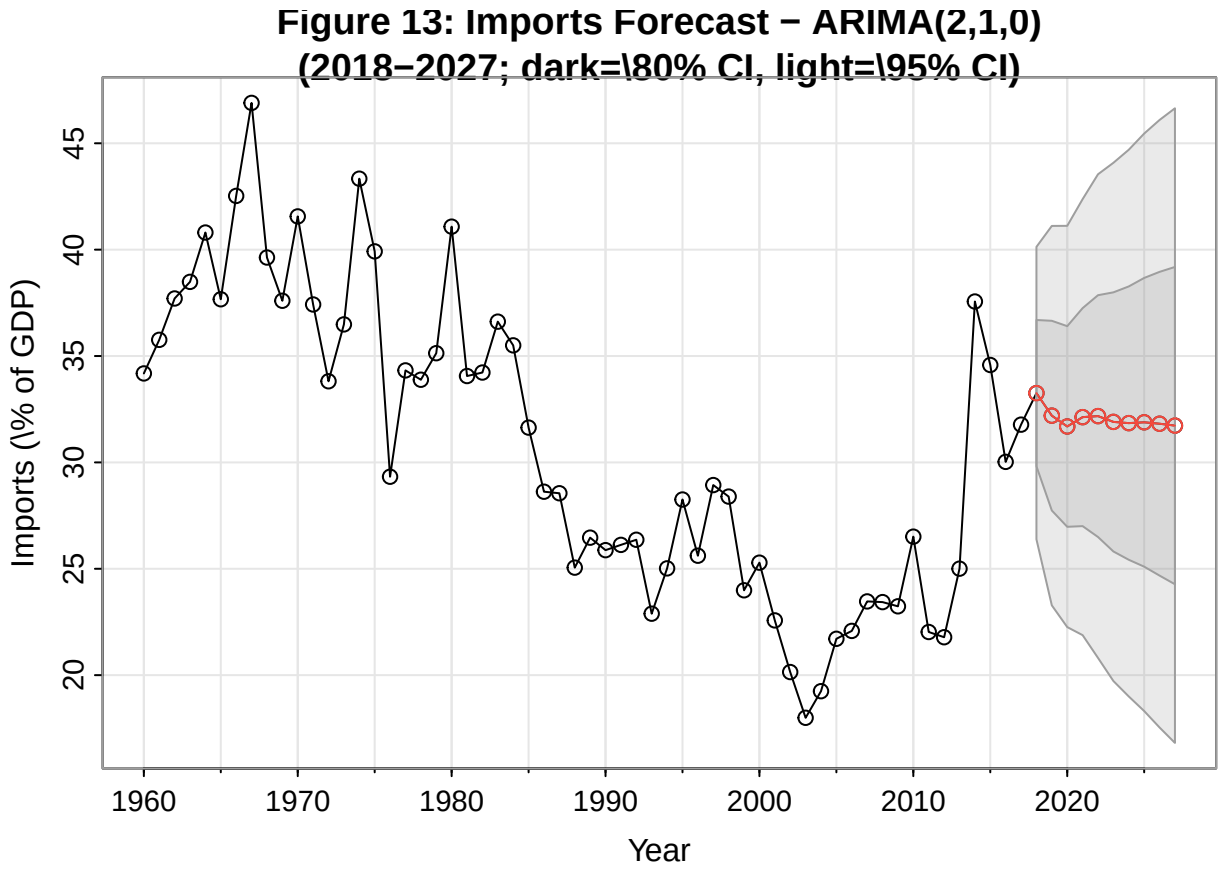


Table 5: Imports Forecasts (% of GDP)

		\80% CI		\95% CI	
Year	Forecast	Lower 80%	Upper 80%	Lower 95%	Upper 95%
2018	33.25692	28.85799	37.65584	26.52106	39.99277
2019	32.19773	26.49032	37.90513	23.45826	40.93720
2020	31.68879	25.65335	37.72423	22.44702	40.93055
2021	32.12781	25.56962	38.68599	22.08558	42.17003
2022	32.17661	24.90183	39.45140	21.03710	43.31613
2023	31.90387	24.10890	39.69885	19.96782	43.83992
2024	31.84707	23.62323	40.07090	19.25431	44.43982
2025	31.88484	23.19897	40.57072	18.58460	45.18509
2026	31.81757	22.67970	40.95545	17.82520	45.80995
2027	31.72968	22.18425	41.27511	17.11325	46.34612

The forecasts for Imports (% of GDP) over the period 2018–2027 are generated from the selected

ARIMA(2,1,0) model.

The point forecasts indicate that Imports remain relatively stable over time, fluctuating slightly around 31–33% of GDP. Unlike GDP, there is no strong upward or downward trend, suggesting that Imports as a share of GDP are expected to remain steady in the long run.

However, both the 80% and 95% prediction intervals widen gradually as the forecast horizon increases. For example, the 95% interval expands from approximately 26.52–39.99 in 2018 to 17.11–46.35 in 2027. This widening reflects increasing uncertainty in future predictions, which is typical in time series forecasting.

Overall, the model suggests that while the central tendency of Imports remains stable, there is considerable uncertainty in long-term forecasts, indicating potential variability in future import behavior.

8 Conclusion

This report analyzed the GDP and Imports (% of GDP) time series of the Central African Republic from 1960 to 2017. A log transformation was applied to GDP to stabilize variance, while Imports required no transformation. ADF tests confirmed both series needed first-order differencing to achieve stationarity. Based on AIC, BIC, and Ljung-Box diagnostics, ARIMA(0,1,0) with drift was selected for log(GDP) and ARIMA(2,1,0) for Imports. GDP is forecast to grow steadily from 2.05 to 3.22 billion USD over 2018-2027, while Imports are expected to remain stable around 31-33% of GDP. CCF analysis further revealed that the two series share a lagged rather than contemporaneous relationship, suggesting that import activity influences GDP growth over the long run.

Appendix: R Code

```r
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)
library(astsa)      # sarima(), sarima.for()
library(forecast)   # Arima(), auto.arima(), forecast()
library(MASS)       # boxcox()
library(tseries)    # adf.test()
library(ggplot2)
library(gridExtra)
load("D:/UC Davis/Course/STA137/finalproject.Rdata")
sapply(finalPro_data,class)
```

```
##    Country       Code       Year        GDP     Growth        CPI    Imports
##   "factor"   "factor"  "numeric"  "numeric"  "numeric"  "numeric"  "numeric"
##    Exports Population
##  "numeric"  "numeric"
```

```r
data_CAR <- finalPro_data[, c("Year", "GDP", "Imports")]
data_CAR <- data_CAR[order(data_CAR$Year), ]
Year    <- as.integer(data_CAR$Year)
GDP     <- data_CAR$GDP        # current USD
Imports <- data_CAR$Imports    # % of GDP
n       <- nrow(data_CAR)      # 58 observations
cat("=== DATA OVERVIEW ===\n")
```

```
## === DATA OVERVIEW ===
```

```r
cat("n =", n, "observations, 1960-2017\n")
```

```
## n = 58 observations, 1960-2017
```

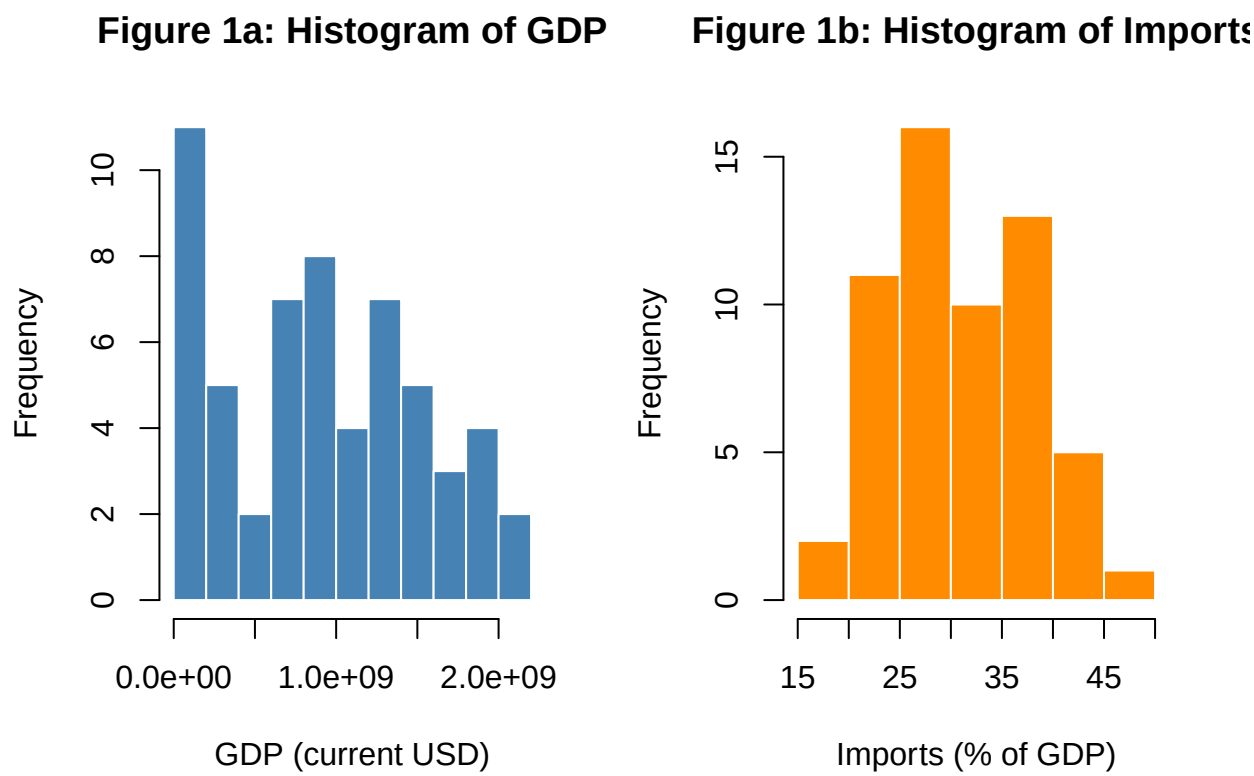
```r
cat("Missing values:", sum(is.na(data_CAR)), "\n\n")
```

```
## Missing values: 0
```

```r
print(summary(data_CAR[, c("GDP","Imports")]))
```

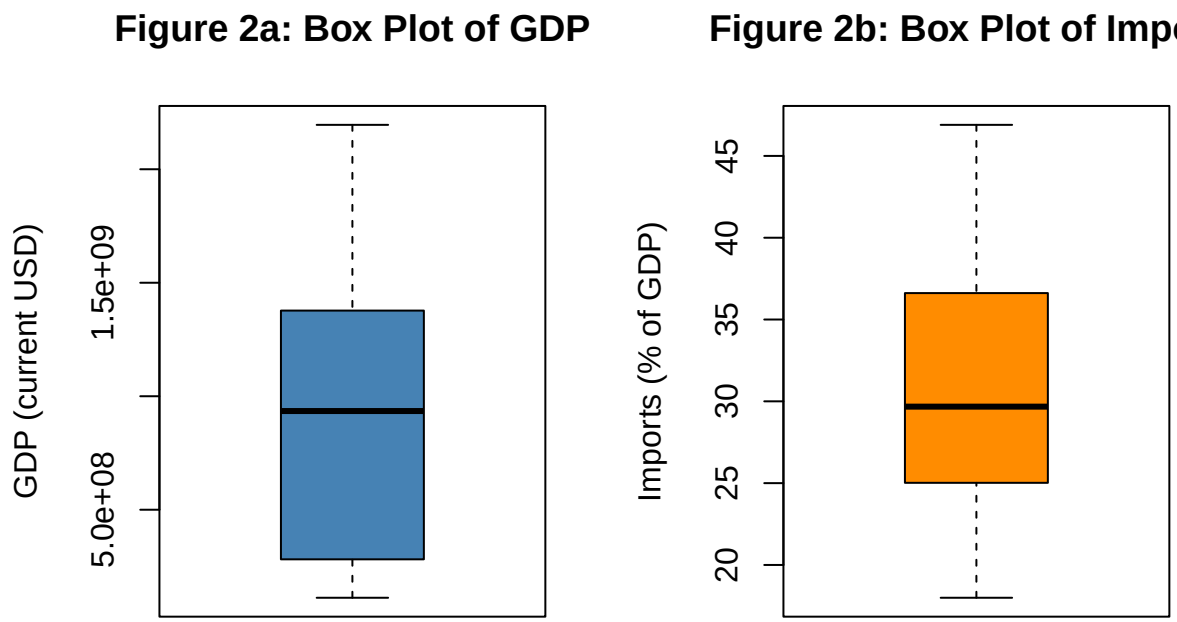
```
##       GDP               Imports
##  Min.   :1.122e+08   Min.   :18.00
##  1st Qu.:3.057e+08   1st Qu.:25.03
##  Median :9.348e+08   Median :29.68
##  Mean   :9.394e+08   Mean   :30.83
##  3rd Qu.:1.367e+09   3rd Qu.:36.58
##  Max.   :2.196e+09   Max.   :46.90
```

```r
#   Figure 1: Histograms
par(mfrow = c(1, 2))

hist(GDP,
     breaks = 10,
     col    = "steelblue",
     border = "white",
     main   = "Figure 1a: Histogram of GDP",
     xlab   = "GDP (current USD)",
     ylab   = "Frequency")

hist(Imports,
     breaks = 10,
     col    = "darkorange",
     border = "white",
     main   = "Figure 1b: Histogram of Imports",
     xlab   = "Imports (% of GDP)",
     ylab   = "Frequency")
```




```r
# Figure 2: Box plots
par(mfrow = c(1, 2))

boxplot(GDP,
        col  = "steelblue",
```

```r
        main = "Figure 2a: Box Plot of GDP",
        ylab = "GDP (current USD)")

boxplot(Imports,
        col  = "darkorange",
        main = "Figure 2b: Box Plot of Imports",
        ylab = "Imports (% of GDP)")
```




```r
#   Figure 3: Time series plots
par(mfrow = c(1, 2))

plot(Year, GDP,
     type = "o", pch = 16, cex = 0.7, col = "steelblue",
     main = "Figure 3a: CAR GDP (1960-2017)",
     xlab = "Year",
     ylab = "GDP (current USD)")

plot(Year, Imports,
     type = "o", pch = 16, cex = 0.7, col = "darkorange",
     main = "Figure 3b: CAR Imports (1960-2017)",
     xlab = "Year",
     ylab = "Imports (% of GDP)")
```

Figure 3a: CAR GDP (1960–2017

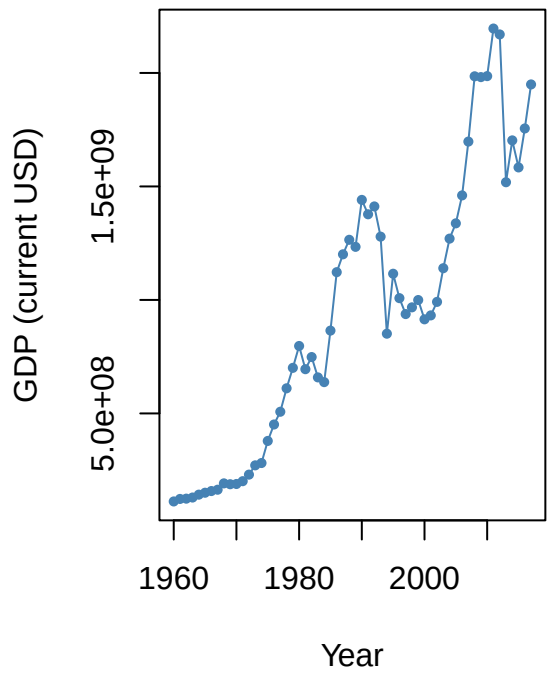
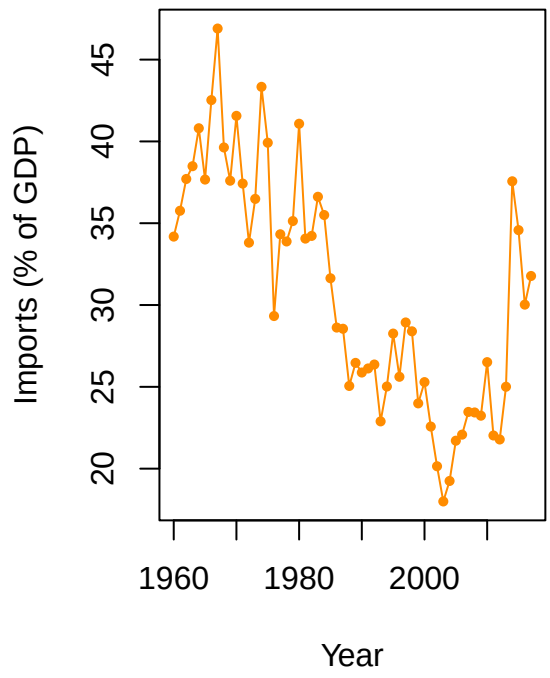
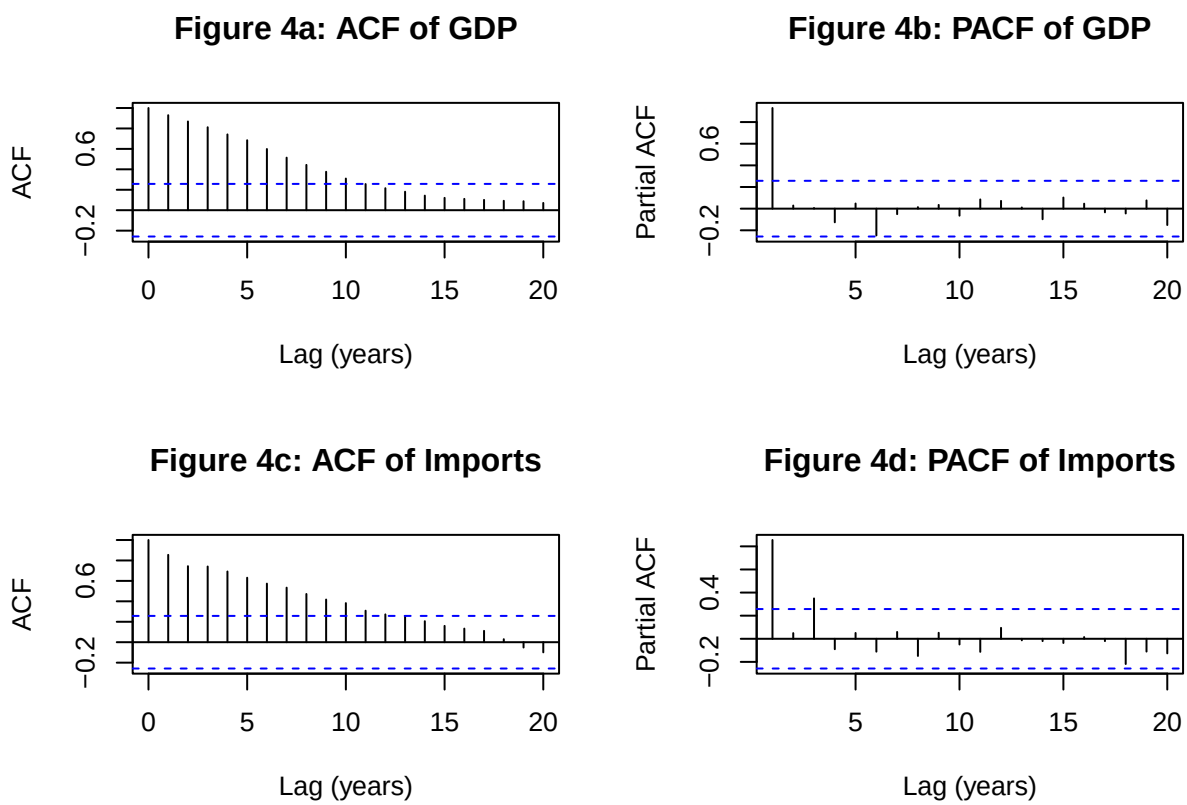


Figure 3b: CAR Imports (1960–20




```
#   Figure 4: ACF and PACF of raw series
par(mfrow = c(2, 2))
acf(GDP,  lag.max=20,
    main="Figure 4a: ACF of GDP",
    xlab="Lag (years)", ylab="ACF")
pacf(GDP, lag.max=20,
     main="Figure 4b: PACF of GDP",
     xlab="Lag (years)", ylab="Partial ACF")
acf(Imports,  lag.max=20,
    main="Figure 4c: ACF of Imports",
    xlab="Lag (years)", ylab="ACF")
pacf(Imports, lag.max=20,
     main="Figure 4d: PACF of Imports",
     xlab="Lag (years)", ylab="Partial ACF")
```




```r
half <- n %/% 2     # split point = observation 29 (year 1988)
# For GDP
var_GDP_1st <- var(GDP[1:half])
var_GDP_2nd <- var(GDP[(half+1):n])
ratio_GDP   <- var_GDP_2nd / var_GDP_1st

cat(sprintf("GDP variance - 1st half: %.3e\n", var_GDP_1st))
```

```
## GDP variance - 1st half: 1.232e+17
```

```r
cat(sprintf("GDP variance - 2nd half: %.3e\n", var_GDP_2nd))
```

```
## GDP variance - 2nd half: 1.696e+17
```

```r
cat(sprintf("Ratio (2nd/1st):         %.3f\n", ratio_GDP))
```

```
## Ratio (2nd/1st):         1.377
```

```r
cat("--> Variance in 2nd half is", round(ratio_GDP,1), "x larger\n")
```

```
## --> Variance in 2nd half is 1.4 x larger
```

```r
cat("--> GDP IS heteroscedastic -> transformation NEEDED\n\n")
```

```
## --> GDP IS heteroscedastic -> transformation NEEDED
```

```r
# For Imports
var_IMP_1st <- var(Imports[1:half])
var_IMP_2nd <- var(Imports[(half+1):n])
ratio_IMP   <- var_IMP_2nd / var_IMP_1st

cat(sprintf("Imports variance - 1st half: %.4f\n", var_IMP_1st))
```

```
## Imports variance - 1st half: 23.3995
```

```r
cat(sprintf("Imports variance - 2nd half: %.4f\n", var_IMP_2nd))
```

```
## Imports variance - 2nd half: 18.7930
```

```r
cat(sprintf("Ratio (2nd/1st):             %.4f\n", ratio_IMP))
```

```
## Ratio (2nd/1st):             0.8031
```

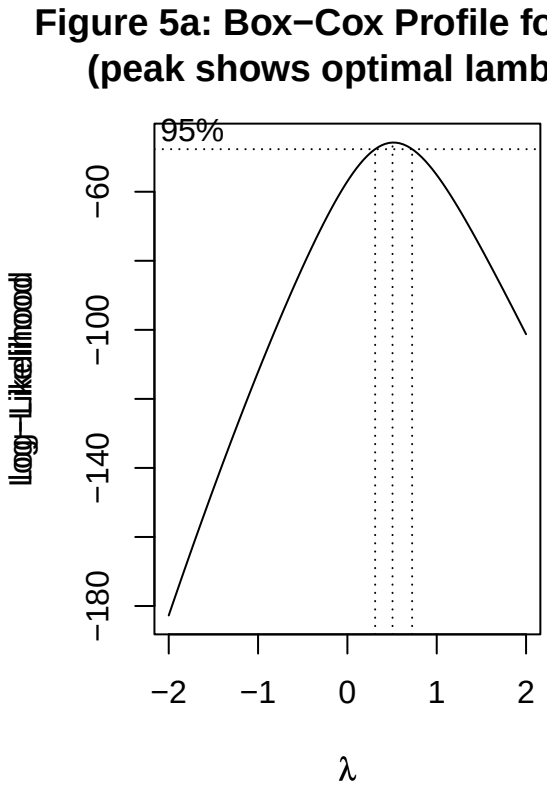
```r
cat("--> Variance in 2nd half is", round(ratio_IMP,2), "x (close to 1.0)\n")
```

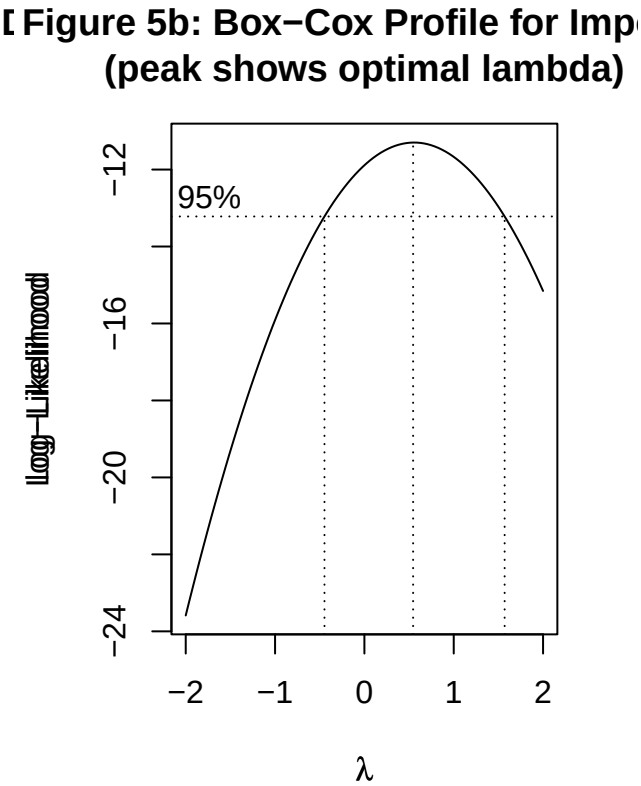
```
## --> Variance in 2nd half is 0.8 x (close to 1.0)
```

```r
cat("--> Imports IS NOT heteroscedastic -> transformation NOT needed\n")
```

```
## --> Imports IS NOT heteroscedastic -> transformation NOT needed
```

```r
par(mfrow = c(1, 2))

bc_GDP <- boxcox(GDP ~ Year, data=data_CAR, lambda=seq(-2, 2, 0.1))
title(main = "Figure 5a: Box-Cox Profile for GDP\n(peak shows optimal lambda)",
      xlab = expression(lambda),
      ylab = "Log-Likelihood")
lambda_GDP <- bc_GDP$x[which.max(bc_GDP$y)]

bc_IMP <- boxcox(Imports ~ Year, data=data_CAR, lambda=seq(-2, 2, 0.1))
title(main = "Figure 5b: Box-Cox Profile for Imports\n(peak shows optimal lambda)",
      xlab = expression(lambda),
      ylab = "Log-Likelihood")
```






```
lambda_IMP <- bc_IMP$x[which.max(bc_IMP$y)]

log_GDP <- log(GDP)

var_logGDP_1st <- var(log_GDP[1:half])
var_logGDP_2nd <- var(log_GDP[(half+1):n])
ratio_logGDP   <- var_logGDP_2nd / var_logGDP_1st
# Also verify on the differenced series - the most critical check
diff_GDP    <- diff(GDP)
diff_logGDP <- diff(log_GDP)
diff_IMP    <- diff(Imports)

var_dGDP_1st  <- var(diff_GDP[1:(half-1)])
var_dGDP_2nd  <- var(diff_GDP[half:length(diff_GDP)])
var_dlGDP_1st <- var(diff_logGDP[1:(half-1)])
var_dlGDP_2nd <- var(diff_logGDP[half:length(diff_logGDP)])
cat("After differencing:\n")
```

```
## After differencing:
```

```
cat(sprintf("  diff(GDP)     ratio (2nd/1st): %.3f  <- variance explodes!\n",
            var_dGDP_2nd/var_dGDP_1st))
```

```
##   diff(GDP)     ratio (2nd/1st): 7.050  <- variance explodes!
```

```r
cat(sprintf("  diff(log(GDP)) ratio (2nd/1st): %.3f  <- much more stable \n",
            var_dlGDP_2nd/var_dlGDP_1st))
```

```
##   diff(log(GDP)) ratio (2nd/1st): 1.795  <- much more stable
```

```r
cat("\nConclusion: diff(log(GDP)) has much more stable variance\n")
```

```
##
## Conclusion: diff(log(GDP)) has much more stable variance
```

```r
cat("than diff(GDP). This is the critical check that confirms\n")
```

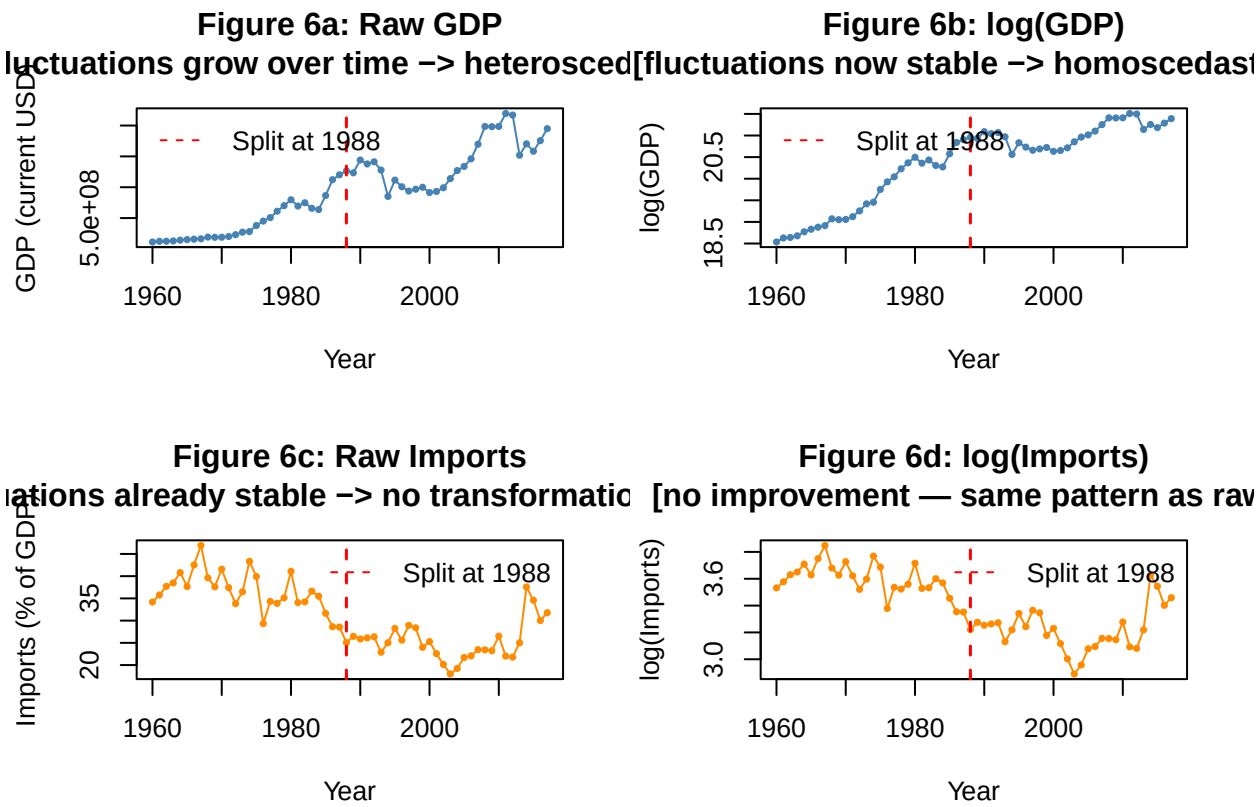
```
## than diff(GDP). This is the critical check that confirms
```

```r
cat("we should use log(GDP) before differencing.\n")
```

```
## we should use log(GDP) before differencing.
```

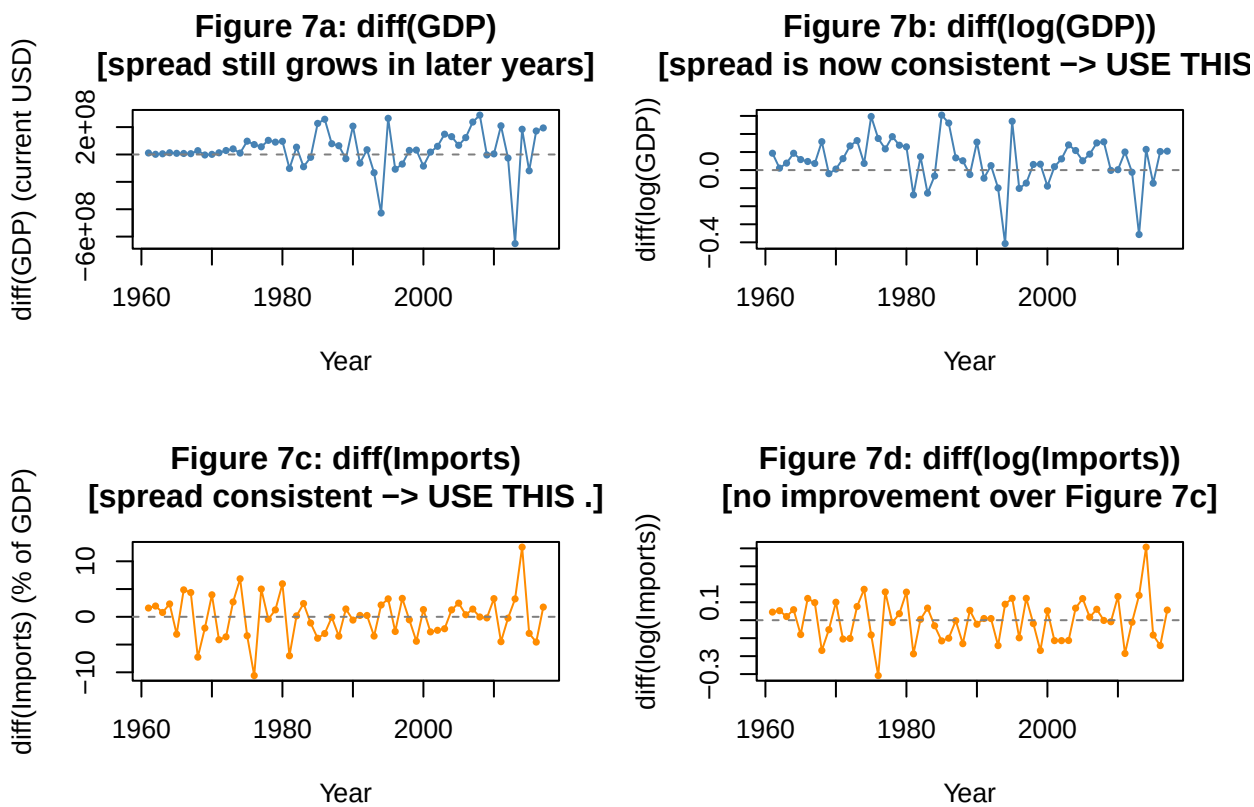
```r
# This figure IS the proof - show both raw and log on same scale
par(mfrow = c(2, 2))

plot(Year, GDP,
     type="o", pch=16, cex=0.6, col="steelblue",
     main="Figure 6a: Raw GDP\n[fluctuations grow over time → heteroscedastic]",
     xlab="Year", ylab="GDP (current USD)")
# Draw a cone to show growing variance
abline(v=Year[half], col="red", lty=2, lwd=1.5)
legend("topleft", bty="n", lty=2, col="red",
       legend=paste("Split at", Year[half]))

plot(Year, log_GDP,
     type="o", pch=16, cex=0.6, col="steelblue",
     main="Figure 6b: log(GDP)\n[fluctuations now stable → homoscedastic  ]",
     xlab="Year", ylab="log(GDP)")
abline(v=Year[half], col="red", lty=2, lwd=1.5)
legend("topleft", bty="n", lty=2, col="red",
       legend=paste("Split at", Year[half]))

plot(Year, Imports,
     type="o", pch=16, cex=0.6, col="darkorange",
     main="Figure 6c: Raw Imports\n[fluctuations already stable → no transformation need
     xlab="Year", ylab="Imports (% of GDP)")
abline(v=Year[half], col="red", lty=2, lwd=1.5)
legend("topright", bty="n", lty=2, col="red",
```

```
       legend=paste("Split at", Year[half]))

plot(Year, log(Imports),
     type="o", pch=16, cex=0.6, col="darkorange",
     main="Figure 6d: log(Imports)\n[no improvement - same pattern as raw  ]",
     xlab="Year", ylab="log(Imports)")
abline(v=Year[half], col="red", lty=2, lwd=1.5)
legend("topright", bty="n", lty=2, col="red",
       legend=paste("Split at", Year[half]))
```




```
#   Figure 7: After differencing - the decisive proof
par(mfrow = c(2, 2))

Year_diff <- Year[-1]

plot(Year_diff, diff_GDP,
     type="o", pch=16, cex=0.6, col="steelblue",
     main="Figure 7a: diff(GDP)\n[spread still grows in later years]",
     xlab="Year", ylab="diff(GDP) (current USD)")
abline(h=0, lty=2, col="gray50")

plot(Year_diff, diff_logGDP,
     type="o", pch=16, cex=0.6, col="steelblue",
     main="Figure 7b: diff(log(GDP))\n[spread is now consistent → USE THIS  ]",
```

```r
     xlab="Year", ylab="diff(log(GDP))")
abline(h=0, lty=2, col="gray50")

plot(Year_diff, diff_IMP,
     type="o", pch=16, cex=0.6, col="darkorange",
     main="Figure 7c: diff(Imports)\n[spread consistent → USE THIS ]",
     xlab="Year", ylab="diff(Imports) (% of GDP)")
abline(h=0, lty=2, col="gray50")

plot(Year_diff, diff(log(Imports)),
     type="o", pch=16, cex=0.6, col="darkorange",
     main="Figure 7d: diff(log(Imports))\n[no improvement over Figure 7c]",
     xlab="Year", ylab="diff(log(Imports))")
abline(h=0, lty=2, col="gray50")
```




```r
library(knitr)
library(kableExtra)
library(knitr)

# Build the data frame
transformation_table <- data.frame(
  Series    = c("GDP", "Imports"),
  `Var Ratio (Raw)`  = c(sprintf("%.2fx (unstable)", ratio_GDP),
                         sprintf("%.2fx (stable)",   ratio_IMP)),
```

```r
  `Var Ratio (Log)`  = c(sprintf("%.2fx (stable)",  ratio_logGDP),
                         sprintf("%.2fx (worse)",   var(log(Imports)[(half+1):n]) /
                                                    var(log(Imports)[1:half]))),
  Decision = c("USE LOG", "NO LOG"),
  check.names = FALSE
)


kable(transformation_table,
      caption = "Transformation Decision Summary",
      align   = c("l", "c", "c", "c"),
      booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position")) |>
  column_spec(4, bold = TRUE,
              color = ifelse(transformation_table$Decision == "USE LOG", "blue", "black'
```

Table 6: Transformation Decision Summary

Series	Var Ratio (Raw)	Var Ratio (Log)	Decision
GDP	1.38x (unstable)	0.13x (stable)	**USE LOG**
Imports	0.80x (stable)	1.41x (worse)	**NO LOG**

```r
adf_table <- data.frame(
  Series          = c("GDP", "log(GDP)", "Imports", "diff(log(GDP))", "diff(Imports)"),
  `ADF Statistic` = c(-3.1953, -1.9149, -1.3131, -3.3099, -4.7438),
  `P-Value`       = c(0.0969,  0.6090,  0.8520,  0.0790,  0.0100),
  Result          = c("Non-stationary", "Non-stationary", "Non-stationary",
                      "Borderline Non-stationary", "Stationary"),  # <-- changed here
  check.names = FALSE
)

kable(adf_table,
      caption  = "ADF Unit Root Test Results",
      align    = c("l", "c", "c", "c"),
      booktabs = TRUE,
      digits   = 4) |>
  kable_styling(latex_options = c("striped", "hold_position")) |>
  column_spec(4,
              bold  = TRUE,
              color = ifelse(adf_table$Result == "Stationary", "blue",
                      ifelse(adf_table$Result == "Borderline Non-stationary", "orange",
                             "black")))  # <-- added orange for borderline
```

Table 7: ADF Unit Root Test Results

Series	ADF Statistic	P-Value	Result
GDP	-3.1953	0.0969	**Non-stationary**
log(GDP)	-1.9149	0.6090	**Non-stationary**
Imports	-1.3131	0.8520	**Non-stationary**
diff(log(GDP))	-3.3099	0.0790	**Borderline Non-stationary**
diff(Imports)	-4.7438	0.0100	**Stationary**

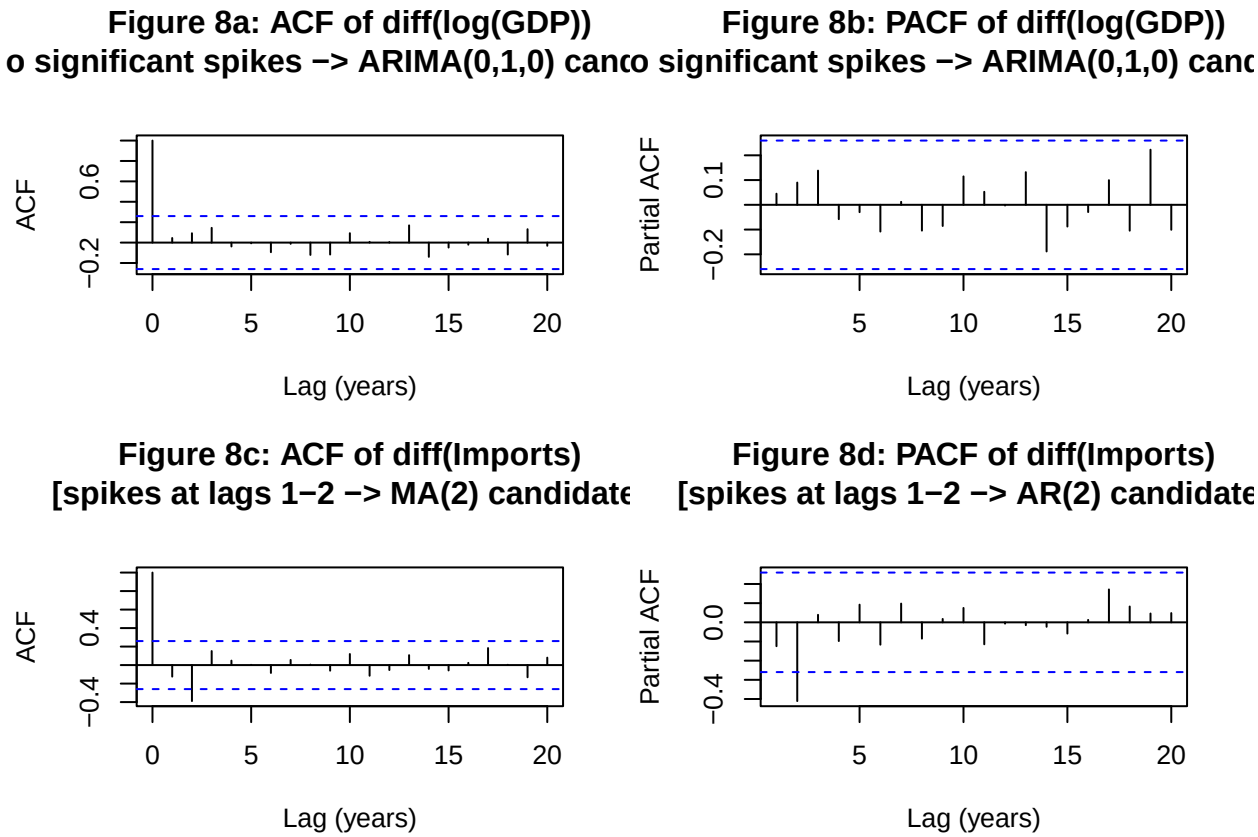
```
conclusion_table <- data.frame(
  Series          = c("log(GDP)", "Imports", "diff(log(GDP))", "diff(Imports)"),
  Status          = c("Non-stationary", "Non-stationary", "Stationary", "Stationary"),
  `Differencing`  = c("Needs d=1", "Needs d=1", "Stationary at d=1", "Stationary at d=1"
  check.names = FALSE
)

kable(conclusion_table,
      caption  = "Stationarity Conclusion \\& Differencing Decision",
      align    = c("l", "c", "c"),
      booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position")) |>
  column_spec(2,
              bold  = TRUE,
              color = ifelse(conclusion_table$Status == "Stationary", "blue", "black"))
```

Table 8: Stationarity Conclusion & Differencing Decision

Series	Status	Differencing
log(GDP)	**Non-stationary**	Needs d=1
Imports	**Non-stationary**	Needs d=1
diff(log(GDP))	**Stationary**	Stationary at d=1
diff(Imports)	**Stationary**	Stationary at d=1

```
#   Figure 8: ACF and PACF after transformation + differencing
par(mfrow = c(2, 2))
acf(diff_logGDP, lag.max=20,
    main="Figure 8a: ACF of diff(log(GDP))\n[no significant spikes -> ARIMA(0,1,0) candi
    xlab="Lag (years)", ylab="ACF")
pacf(diff_logGDP, lag.max=20,
     main="Figure 8b: PACF of diff(log(GDP))\n[no significant spikes -> ARIMA(0,1,0) can
     xlab="Lag (years)", ylab="Partial ACF")
acf(diff_IMP, lag.max=20,
    main="Figure 8c: ACF of diff(Imports)\n[spikes at lags 1-2 -> MA(2) candidate]",
    xlab="Lag (years)", ylab="ACF")
pacf(diff_IMP, lag.max=20,
```

```
     main="Figure 8d: PACF of diff(Imports)\n[spikes at lags 1-2 -> AR(2) candidate]",
     xlab="Lag (years)", ylab="Partial ACF")
```




```
cat("\n=== CROSS-CORRELATION ANALYSIS ===\n")
```

```
##
## === CROSS-CORRELATION ANALYSIS ===
```

```
cat("Series used: diff(log(GDP)) and diff(Imports)\n")
```

```
## Series used: diff(log(GDP)) and diff(Imports)
```

```
cat("Both are stationary. Positive lag k = GDP leads Imports.\n\n")
```

```
## Both are stationary. Positive lag k = GDP leads Imports.
```

```
ccf_obj <- ccf(diff_logGDP, diff_IMP,
               lag.max=15,
               main="Figure 9a: CCF - diff(log(GDP)) vs diff(Imports)\n(+lag = GDP leads
               xlab="Lag k (years)",
               ylab="Cross-Correlation")
```

```r
cor_res <- cor.test(diff_logGDP, diff_IMP)
plot(diff_logGDP, diff_IMP,
     pch=16, cex=0.9, col="steelblue",
     main="Figure 9b: Scatter - diff(log(GDP)) vs diff(Imports)\n(lag = 0, contemporaneo
     xlab="diff(log(GDP))", ylab="diff(Imports) (% of GDP)")
abline(lm(diff_IMP ~ diff_logGDP), col="red", lwd=2)
legend("topright", bty="n",
       legend=sprintf("r = %.3f,  p = %.3f",
                      cor_res$estimate, cor_res$p.value))

nd       <- length(diff_logGDP)
ci_bound <- 1.96 / sqrt(nd)
cat(sprintf("95%% CI = +/- %.4f  (n = %d)\n\n", ci_bound, nd))
```

```
## 95% CI = +/- 0.2596  (n = 57)
```

```r
ccf_lags <- as.integer(ccf_obj$lag)
ccf_vals <- as.numeric(ccf_obj$acf)
cat("Significant lags (|r| > CI):\n")
```

```
## Significant lags (|r| > CI):
```

```r
for (i in seq_along(ccf_lags)) {
  if (abs(ccf_vals[i]) > ci_bound)
    cat(sprintf("  Lag %+3d:  r = %+.4f  ***\n", ccf_lags[i], ccf_vals[i]))
}
```

```
##   Lag  -1:  r = -0.2937  ***
##   Lag +12:  r = +0.2686  ***
```

```r
cat(sprintf("\nLag 0 (contemporaneous): r = %.4f, p = %.4f -> %s\n",
            cor_res$estimate, cor_res$p.value,
            ifelse(cor_res$p.value<0.05, "SIGNIFICANT","NOT significant")))
```

```
##
## Lag 0 (contemporaneous): r = -0.1257, p = 0.3515 -> NOT significant
```

```r
gdp_orders <- list(c(1,1,0), c(0,1,1), c(2,1,2), c(0,1,0))
gdp_table  <- data.frame(Model=character(), AIC=numeric(),
                         BIC=numeric(), LjungBox_p=numeric(),
                         stringsAsFactors=FALSE)
for (ord in gdp_orders) {
  tryCatch({
    m  <- arima(log_GDP, order=ord)
```

```r
    sm <- sarima(log_GDP, ord[1], ord[2], ord[3], details=FALSE)
    lb <- Box.test(m$residuals, lag=10, type="Ljung-Box")$p.value
    gdp_table <- rbind(gdp_table, data.frame(
      Model      = sprintf("ARIMA(%d,%d,%d)", ord[1],ord[2],ord[3]),
      AIC        = round(sm$ICs["AIC"],4),
      BIC        = round(sm$ICs["BIC"],4),
      LjungBox_p = round(lb,4)
    ))
  }, error=function(e) NULL)
}
```

```
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate     SE t.value p.value
## ar1        0.0444 0.1315  0.3380  0.7367
## constant   0.0502 0.0178  2.8207  0.0067
##
## sigma^2 estimated as 0.01649082 on 55 degrees of freedom
##
## AIC = -1.161776  AICc = -1.157878  BIC = -1.054247
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate     SE t.value p.value
## ma1        0.0382 0.1228  0.3113  0.7568
## constant   0.0502 0.0177  2.8416  0.0063
##
## sigma^2 estimated as 0.01649574 on 55 degrees of freedom
##
## AIC = -1.161487  AICc = -1.157589  BIC = -1.053958
##
```

```
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate     SE  t.value p.value
## ar1        1.5232 0.1236  12.3224  0.0000
## ar2       -0.9044 0.0924  -9.7839  0.0000
## ma1       -1.5297 0.1006 -15.2086  0.0000
## ma2        1.0000 0.0999  10.0098  0.0000
## constant   0.0508 0.0196   2.5979  0.0122
##
## sigma^2 estimated as 0.01443687 on 52 degrees of freedom
##
## AIC = -1.138155  AICc = -1.117515  BIC = -0.9230966
```

```
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate    SE t.value p.value
## constant   0.0501 0.017  2.9421  0.0047
##
## sigma^2 estimated as 0.01652443 on 56 degrees of freedom
##
## AIC = -1.194863  AICc = -1.193587  BIC = -1.123177
##
```

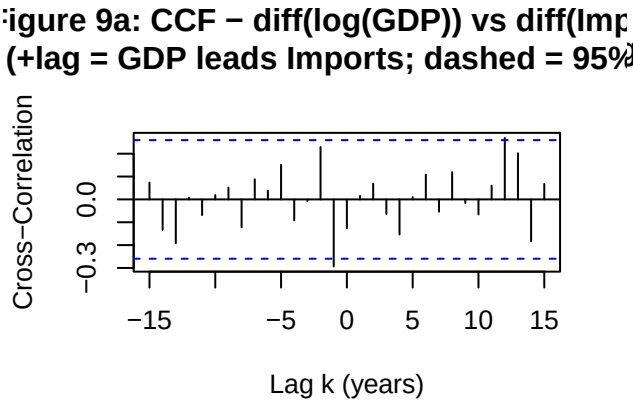
```r
gdp_table$Selected <- ifelse(gdp_table$AIC==min(gdp_table$AIC),"<-- BEST","")
cat("GDP Model Comparison:\n")
```

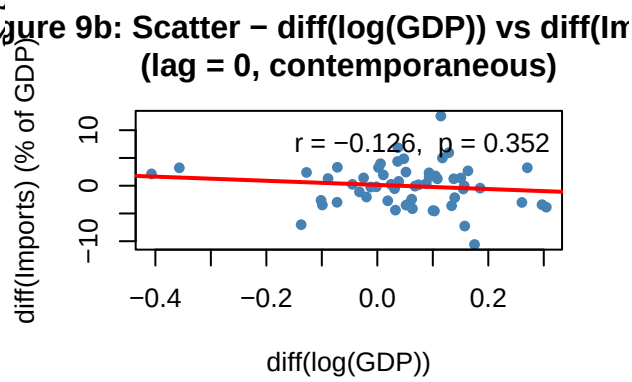
```
## GDP Model Comparison:
```

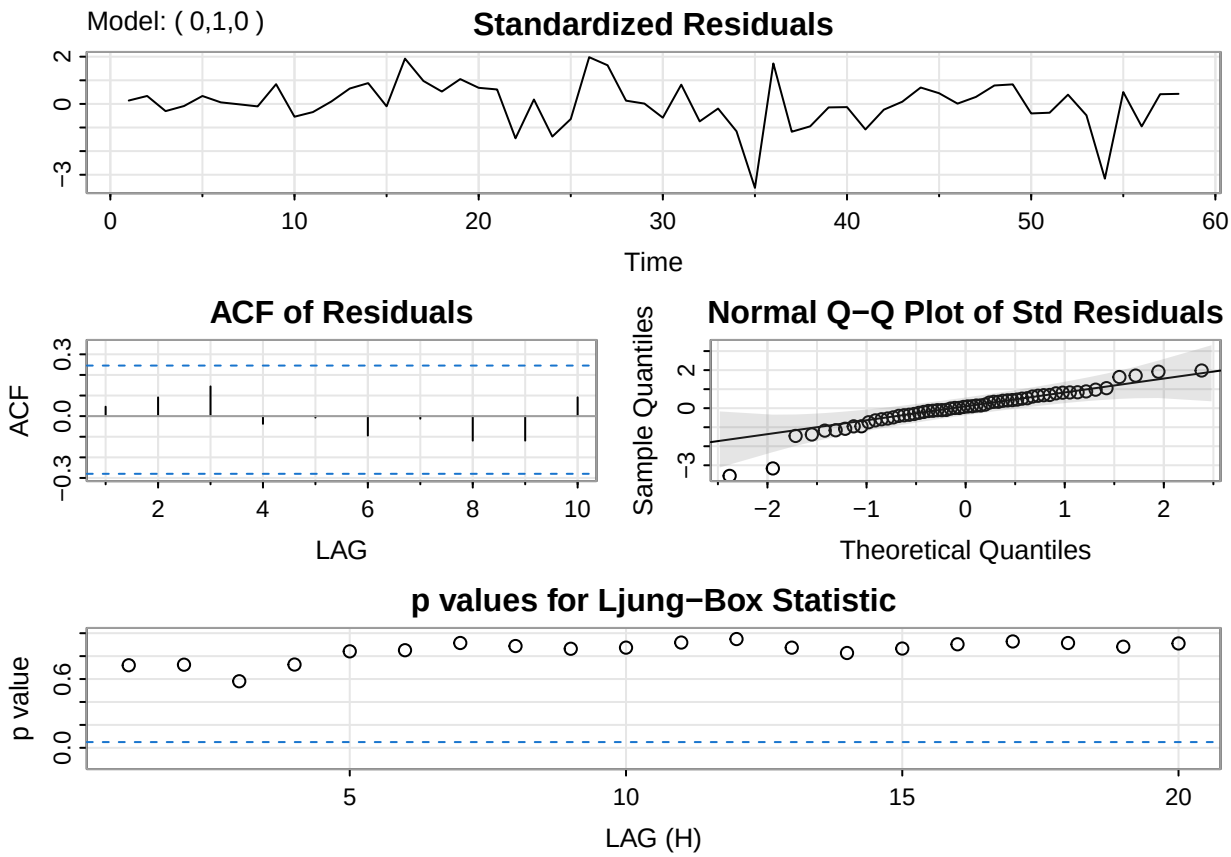
```r
print(gdp_table, row.names=FALSE)
```

```
##         Model     AIC     BIC LjungBox_p Selected
##  ARIMA(1,1,0) -1.1618 -1.0542     0.8166
##  ARIMA(0,1,1) -1.1615 -1.0540     0.8521
##  ARIMA(2,1,2) -1.1382 -0.9231     0.9363
##  ARIMA(0,1,0) -1.1949 -1.1232     0.8684 <-- BEST
```

```r
gdp_final <- arima(log_GDP, order = c(0,1,0))

invisible(
  capture.output(
    suppressMessages(
      gdp_sarima <- sarima(log_GDP, 0, 1, 0)
    )
  )
)
```


igure 9a: CCF − diff(log(GDP)) vs diff(Imp
(+lag = GDP leads Imports; dashed = 95%
Cross−Correlation
0.0
−0.3
−15
−5
0
5
10
15
Lag k (years)


ure 9b: Scatter − diff(log(GDP)) vs diff(In
(lag = 0, contemporaneous)
r = −0.126, p = 0.352
diff(Imports) (% of GDP)
10
0
−10
−0.4
−0.2
0.0
0.2
diff(log(GDP))


Model: (0,1,0)
Standardized Residuals
2
0
−3
0
10
20
30
40
50
60
Time
ACF of Residuals
ACF
0.3
0.0
−0.3
2
4
6
8
10
LAG
Normal Q−Q Plot of Std Residuals
Sample Quantiles
2
0
−3
−2
−1
0
1
2
Theoretical Quantiles
p values for Ljung−Box Statistic
p value
0.6
0.0
5
10
15
20
LAG (H)

```r
lb_gdp <- Box.test(gdp_final$residuals, lag = 10, type = "Ljung-Box")


imp_orders <- list(c(0,1,2), c(2,1,0))
imp_table  <- data.frame(Model=character(), AIC=numeric(),
                         BIC=numeric(), LjungBox_p=numeric(),
                         stringsAsFactors=FALSE)
for (ord in imp_orders) {
  tryCatch({
    m  <- arima(Imports, order=ord)
    sm <- sarima(Imports, ord[1], ord[2], ord[3], details=FALSE)
    lb <- Box.test(m$residuals, lag=10, type="Ljung-Box")$p.value
    imp_table <- rbind(imp_table, data.frame(
      Model      = sprintf("ARIMA(%d,%d,%d)", ord[1],ord[2],ord[3]),
      AIC        = round(sm$ICs["AIC"],4),
      BIC        = round(sm$ICs["BIC"],4),
      LjungBox_p = round(lb,4)
    ))
  }, error=function(e) NULL)
}
```

```
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate     SE t.value p.value
## ma1       -0.0528 0.1300 -0.4065  0.6860
## ma2       -0.4622 0.1416 -3.2636  0.0019
## constant  -0.1157 0.2333 -0.4960  0.6219
##
## sigma^2 estimated as 11.84284 on 54 degrees of freedom
##
## AIC = 5.458552  AICc = 5.466496  BIC = 5.601924
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate     SE t.value p.value
## ar1       -0.1733 0.1203 -1.4410  0.1554
## ar2       -0.4105 0.1196 -3.4317  0.0012
## constant  -0.0531 0.2906 -0.1826  0.8558
##
## sigma^2 estimated as 11.81063 on 54 degrees of freedom
##
## AIC = 5.453971  AICc = 5.461916  BIC = 5.597343
##
```

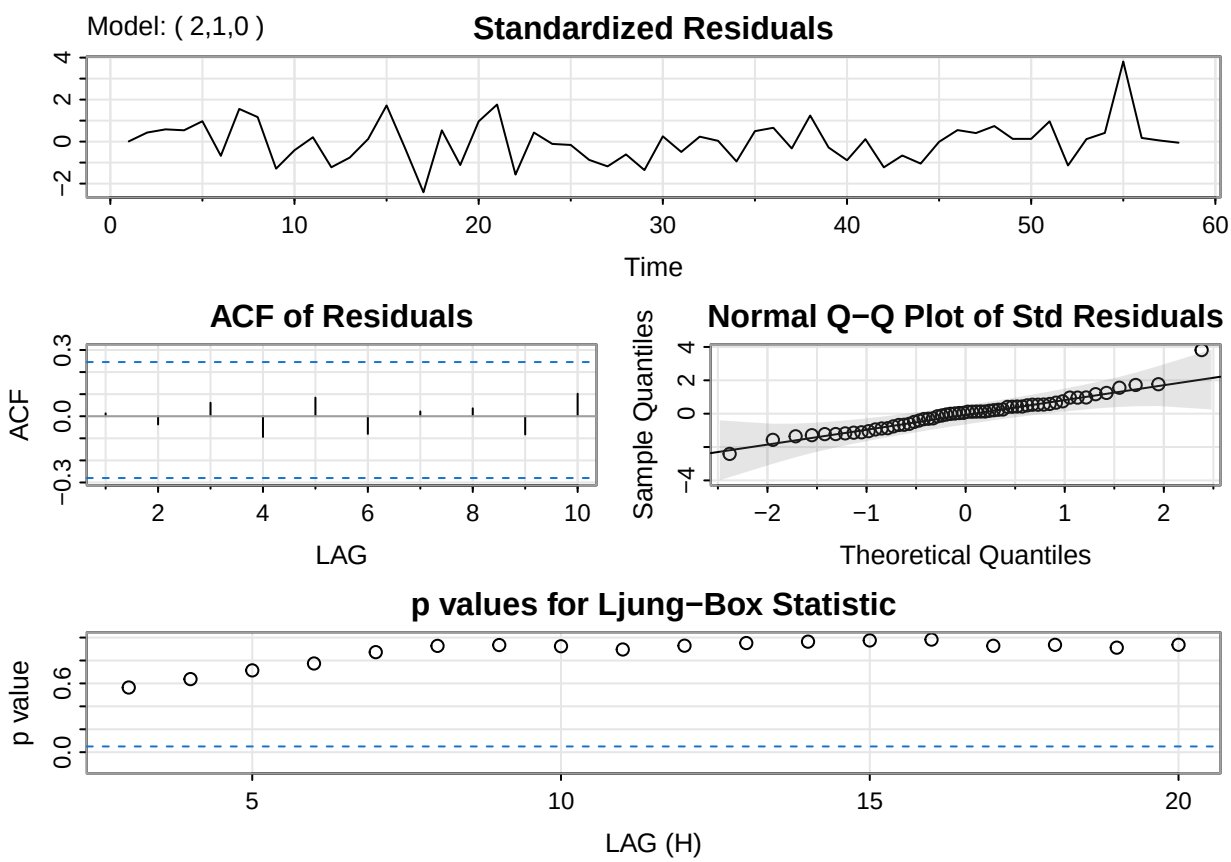
```r
imp_table$Selected <- ifelse(imp_table$AIC==min(imp_table$AIC),"<-- BEST","")
cat("Imports Model Comparison:\n")
```

```
## Imports Model Comparison:
```

```r
print(imp_table, row.names=FALSE)
```

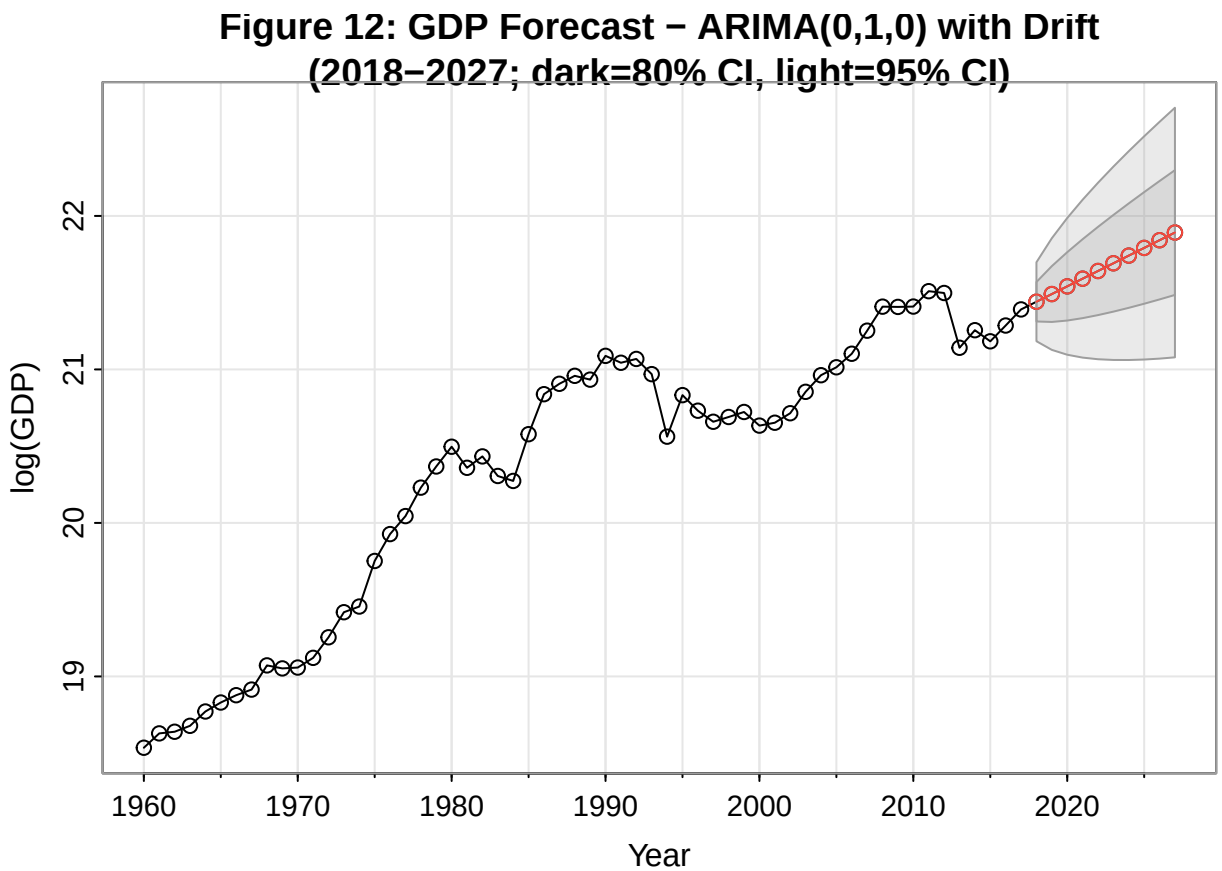
```
##          Model    AIC    BIC LjungBox_p Selected
##   ARIMA(0,1,2) 5.4586 5.6019     0.9363
##   ARIMA(2,1,0) 5.4540 5.5973     0.9779 <-- BEST
```

```r
imp_final  <- arima(Imports, order = c(2, 1, 0))
invisible(
  capture.output(
    suppressMessages(imp_sarima <- sarima(Imports, 2, 1, 0)
)
)
)
```




```r
lb_imp <- Box.test(imp_final$residuals, lag = 10, type = "Ljung-Box")

library(forecast)   # needed for Arima() and autoplot()
```

```r
h <- 10

# Convert to ts objects FIRST - this is the fix
log_GDP_ts <- ts(log_GDP,  start = 1960, frequency = 1)
Imports_ts <- ts(Imports,  start = 1960, frequency = 1)


#   Figure 12: GDP forecast
# Option A: sarima.for() - now works because input is ts object
par(mfrow = c(1, 1))
gdp_fc <- sarima.for(log_GDP_ts,
                     n.ahead = h,
                     p = 0, d = 1, q = 0,
                     main = paste0("Figure 12: GDP Forecast",
                                   " - ARIMA(0,1,0) with Drift\n",
                                   "(2018-2027; dark=80% CI, light=95% CI)"),
                     xlab = "Year",
                     ylab = "log(GDP)")
```



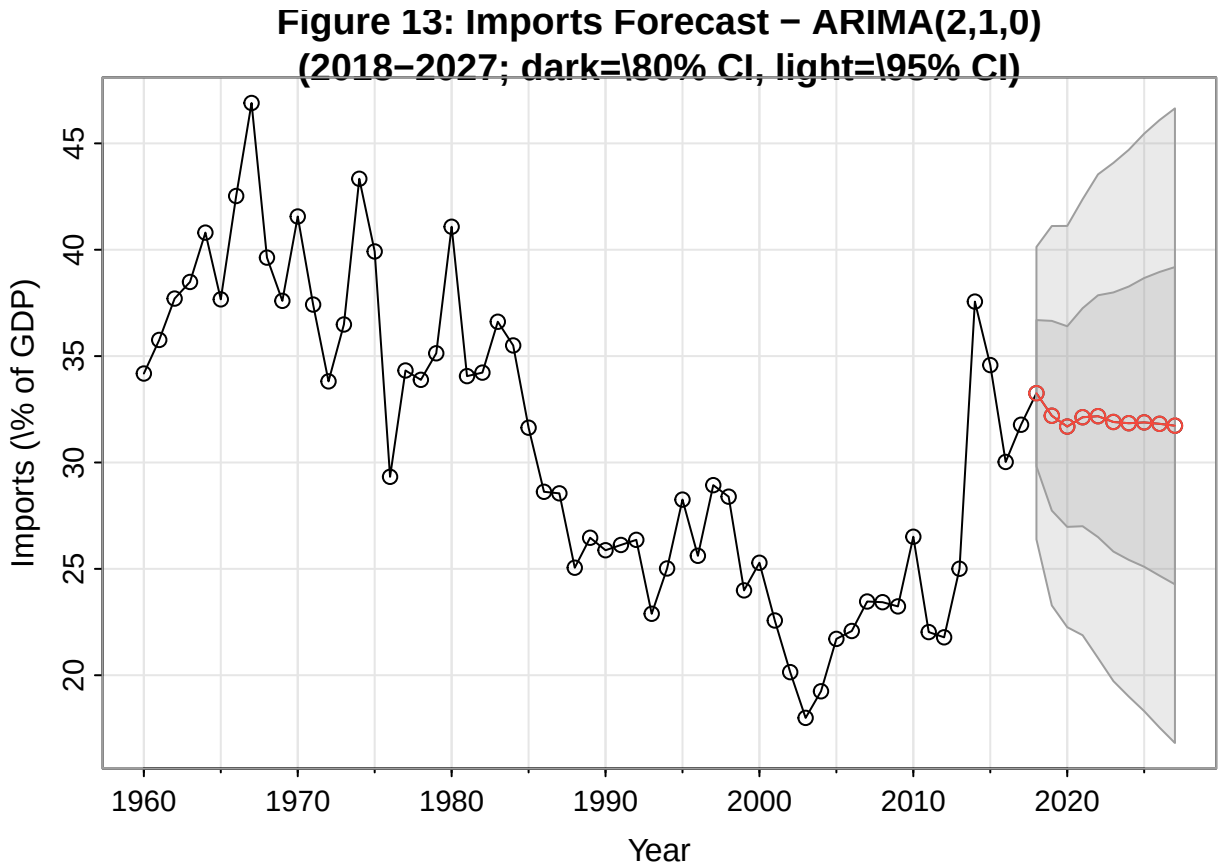

```r
# Suppress all cat() output
gdp_forecast_table <- data.frame(
  Year        = 2018:2027,
  `Forecast`  = sapply(1:h, function(i) exp(gdp_fc$pred[i]) / 1e9),
```

```r
  `Lower 95%` = sapply(1:h, function(i) exp(gdp_fc$pred[i] - 1.96 * gdp_fc$se[i]) / 1e9)
  `Upper 95%` = sapply(1:h, function(i) exp(gdp_fc$pred[i] + 1.96 * gdp_fc$se[i]) / 1e9)
  check.names = FALSE
)

kable(gdp_forecast_table,
      caption  = "GDP Forecasts (Billion USD) - ARIMA(0,1,0) with Drift",
      align    = c("c", "c", "c", "c"),
      booktabs = TRUE,
      digits   = 3) |>
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 9: GDP Forecasts (Billion USD) - ARIMA(0,1,0) with Drift

Year	Forecast	Lower 95%	Upper 95%
2018	2.050	1.593	2.637
2019	2.155	1.509	3.077
2020	2.266	1.464	3.505
2021	2.382	1.439	3.942
2022	2.504	1.426	4.399
2023	2.633	1.420	4.881
2024	2.768	1.421	5.391
2025	2.910	1.427	5.935
2026	3.060	1.437	6.516
2027	3.217	1.450	7.136

```r
#   Figure 13: Imports forecast
par(mfrow = c(1, 1))
imp_fc <- sarima.for(Imports_ts,
                     n.ahead = h,
                     p = 2, d = 1, q = 0,
                     main = paste0("Figure 13: Imports Forecast",
                                   " - ARIMA(2,1,0)\n",
                                   "(2018-2027; dark=\\80% CI, light=\\95% CI)"),
                     xlab = "Year",
                     ylab = "Imports (\\% of GDP)")
```




```r
# Build table from imp_fc directly
imp_forecast_table <- data.frame(
  Year        = 2018:2027,
  `Forecast`  = imp_fc$pred,
  `Lower 80%` = imp_fc$pred - 1.28 * imp_fc$se,
  `Upper 80%` = imp_fc$pred + 1.28 * imp_fc$se,
  `Lower 95%` = imp_fc$pred - 1.96 * imp_fc$se,
  `Upper 95%` = imp_fc$pred + 1.96 * imp_fc$se,
  check.names = FALSE
)

kable(imp_forecast_table,
      caption  = "Imports Forecasts (\\% of GDP)",
      align    = c("c", "c", "c", "c", "c", "c"),
      booktabs = TRUE,
      digits   = 2) |>
  kable_styling(latex_options = c("striped", "hold_position")) |>
  add_header_above(c(" " = 2, "\\80% CI" = 2, "\\95% CI" = 2))
```

Table 10: Imports Forecasts (% of GDP)

		\80% CI		\95% CI	
Year	Forecast	Lower 80%	Upper 80%	Lower 95%	Upper 95%

2018	33.25692	28.85799	37.65584	26.52106	39.99277
2019	32.19773	26.49032	37.90513	23.45826	40.93720
2020	31.68879	25.65335	37.72423	22.44702	40.93055
2021	32.12781	25.56962	38.68599	22.08558	42.17003
2022	32.17661	24.90183	39.45140	21.03710	43.31613
2023	31.90387	24.10890	39.69885	19.96782	43.83992
2024	31.84707	23.62323	40.07090	19.25431	44.43982
2025	31.88484	23.19897	40.57072	18.58460	45.18509
2026	31.81757	22.67970	40.95545	17.82520	45.80995
2027	31.72968	22.18425	41.27511	17.11325	46.34612